5 Databases (284)

ER-model. Relational model:Relational algebra, Tuple calculus, SQL. Integrity constraints, Normal forms. File organization, Indexing (e.g., B and B+ trees). Transactions and concurrency control.

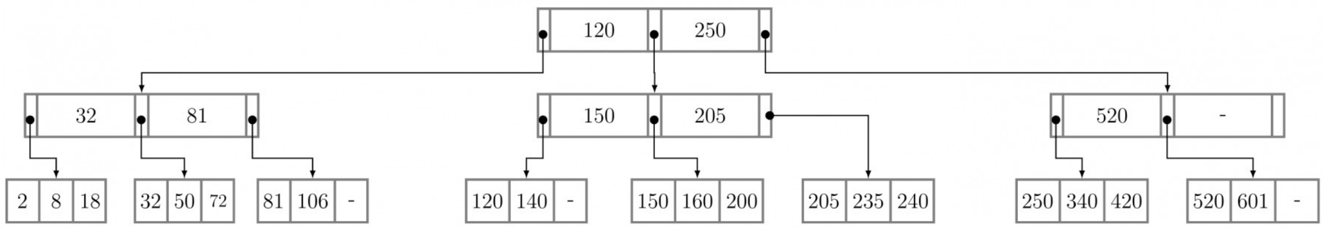
Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximu
1 Mark Count	1	1	4	4	1	3	2	1	1	2.13	4
2 Marks Count	3	4	2	2	2	2	3	3	2	2.63	4
Total Marks	8	9	8	8	5	7	8	7	**5**	**7.5**	**9**

5.1 B Tree (31)

5.1.1 B Tree: GATE CSE 1989 | Question: 12a

The below figure shows a B^+ tree where only key values are indicated in the records. Each block can hold upto three records. A record with a key value 34 is inserted into the B^+ tree. Obtain the modified B^+ tree after insertion.



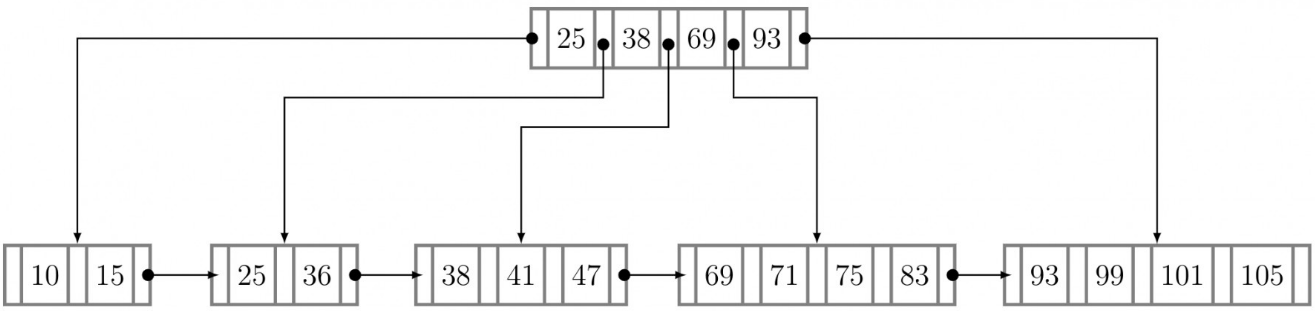
descriptive gate1989 databases b-tree

Answer key

5.1.2 B Tree: GATE CSE 1994 | Question: 14a

Consider B^+ - tree of order d shown in figure. (A B^+ - tree of order d contains between d and $2d$ keys in each node)

Draw the resulting B^+ - tree after 100 is inserted in the figure below.



gate1994 databases b-tree normal descriptive

Answer key

5.1.3 B Tree: GATE CSE 1994 | Question: 14b

For a B^+ - tree of order d with n leaf nodes, the number of nodes accessed during a search is $O(\_)$.

gate1994 databases b-tree normal descriptive

Answer key

5.1.4 B Tree: GATE CSE 1997 | Question: 19

A B^+ - tree of order d is a tree in which each internal node has between d and $2d$ key values. An internal node with M key values has $M+1$ children. The root (if it is an internal node) has between 1 and $2d$ key values. The distance of a node from the root is the length of the path from the root to the node. All leaves are at the same distance from the root. The height of the tree is the distance of a leaf from the root.

A. What is the total number of key values in the internal nodes of a B^+-tree with l leaves ($l \geq 2$)?
B. What is the maximum number of internal nodes in a B^+ - tree of order 4 with 52 leaves?
C. What is the minimum number of leaves in a B^+-tree of order d and height $h (h \geq 1)$?

gate1997 databases b-tree normal descriptive

Answer key

5.1.5 B Tree: GATE CSE 1999 | Question: 1.25

Which of the following is correct?

A. B-trees are for storing data on disk and B^+ trees are for main memory.
B. Range queries are faster on B^+ trees.
C. B-trees are for primary indexes and B^+ trees are for secondary indexes.
D. The height of a B^+ tree is independent of the number of records.

gate1999 databases b-tree normal

Answer key

5.1.6 B Tree: GATE CSE 1999 | Question: 21

Consider a B-tree with degree m, that is, the number of children, c, of any internal node (except the root) is such that $m \leq c \leq 2m-1$. Derive the maximum and minimum number of records in the leaf nodes for such a B-tree with height $h, h \geq 1.$(Assume that the root of a tree is at height 0).

gate1999 databases b-tree normal descriptive

Answer key

5.1.7 B Tree: GATE CSE 2000 | Question: 1.22, UGCNET-June2012-II: 11

B^+-trees are preferred to binary trees in databases because

A. Disk capacities are greater than memory capacities
B. Disk access is much slower than memory access
C. Disk data transfer rates are much less than memory data transfer rates
D. Disks are more reliable than memory

gatecse-2000 databases b-tree normal ugcnetcse-june2012-paper2

Answer key

5.1.8 B Tree: GATE CSE 2000 | Question: 21

(a) Suppose you are given an empty B^+ tree where each node (leaf and internal) can store up to 5 key values. Suppose values $1, 2, \dots 10$ are inserted, in order, into the tree. Show the tree pictorially

i. after 6 insertions, and
ii. after all 10 insertions

Do NOT show intermediate stages.

(b) Suppose instead of splitting a node when it is full, we try to move a value to the left sibling. If there is no left sibling, or the left sibling is full, we split the node. Show the tree after values $1, 2, \dots, 9$ have been inserted. Assume, as in (a) that each node can hold up to 5 keys.

(c) In general, suppose a B^+ tree node can hold a maximum of m keys,and you insert a long sequence of keys in increasing order. Then what approximately is the average number of keys in each leaf level node.

i. in the normal case, and
ii. with the insertion as in (b).

gatecse-2000 databases b-tree normal descriptive

Answer key

5.1.9 B Tree: GATE CSE 2001 | Question: 22

We wish to construct a B^+ tree with fan-out (the number of pointers per node) equal to 3 for the following set of key values:

$80, 50, 10, 70, 30, 100, 90$

Assume that the tree is initially empty and the values are added in the order given.

a. Show the tree after insertion of 10, after insertion of 30, and after insertion of 90. Intermediate trees need not be shown.
b. The key values 30 and 10 are now deleted from the tree in that order show the tree after each deletion.

gatecse-2001 databases b-tree normal descriptive

Answer key

5.1.10 B Tree: GATE CSE 2002 | Question: 17

a. The following table refers to search items for a key in B-trees and B^+ trees.

B-tree		**B^+-tree**	
Successful search	Unsuccessful search	Successful search	Unsuccessful search
X_1	X_2	X_3	X_4

A successful search means that the key exists in the database and unsuccessful means that it is not present in the database. Each of the entries X_1, X_2, X_3 and X_4 can have a value of either Constant or Variable. Constant means that the search time is the same, independent of the specific key value, where variable means that it is dependent on the specific key value chosen for the search.
Give the correct values for the entries X_1, X_2, X_3 and X_4 (for example $X_1 = \text{Constant}$, $X_2 = \text{Constant}$, $X_3 = \text{Constant}$, $X_4 = \text{Constant}$)

b. Relation $R(A, B)$ has the following view defined on it:

```
CREATE VIEW V AS
(SELECT R1.A,R2.B
FROM R AS R1, R as R2
WHERE R1.B=R2.A)
```

i. The current contents of relation R are shown below. What are the contents of the view V?

A	B
1	2
2	3
2	4
4	5
6	7
6	8
9	10

ii. The tuples $(2, 11)$ and $(11, 6)$ are now inserted into R. What are the *additional* tuples that are inserted in V?

gatecse-2002 databases b-tree normal descriptive

Answer key

5.1.11 B Tree: GATE CSE 2002 | Question: 2.23, UGCNET-June2012-II: 26

A B^+ - tree index is to be built on the *Name* attribute of the relation *STUDENT.* Assume that all the student names are of length 8 bytes, disk blocks are of size 512 bytes, and index pointers are of size 4 bytes. Given the scenario, what would be the best choice of the degree (i.e. number of pointers per node) of the B^+ - tree?

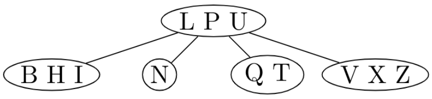
A. 16
B. 42
C. 43
D. 44

gatecse-2002 databases b-tree normal ugcnetcse-june2012-paper2

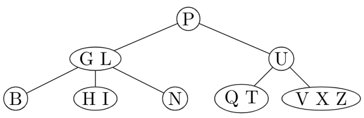
Answer key

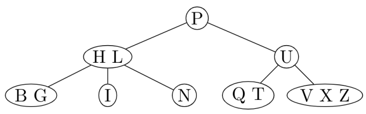
5.1.12 B Tree: GATE CSE 2003 | Question: 65

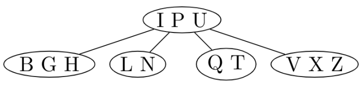
Consider the following $2-3-4$ tree (i.e., B-tree with a minimum degree of two) in which each data item is a letter. The usual alphabetical ordering of letters is used in constructing the tree.



What is the result of inserting G in the above tree?

A. 

B. 

C. 

D. None of the above

gatecse-2003 databases b-tree normal

Answer key

5.1.13 B Tree: GATE CSE 2004 | Question: 52

The order of an internal node in a $B+$ tree index is the maximum number of children it can have. Suppose that a child pointer takes 6 bytes, the search field value takes 14 bytes, and the block size is 512 bytes. What is the order of the internal node?

A. 24
B. 25
C. 26
D. 27

gatecse-2004 databases b-tree normal

Answer key

5.1.14 B Tree: GATE CSE 2005 | Question: 28

Which of the following is a key factor for preferring B^+-trees to binary search trees for indexing database relations?

A. Database relations have a large number of records
B. Database relations are sorted on the primary key
C. B^+-trees require less memory than binary search trees
D. Data transfer from disks is in blocks

gatecse-2005 databases b-tree normal

Answer key

5.1.15 B Tree: GATE CSE 2007 | Question: 63, ISRO2016-59

The order of a leaf node in a B^+ - tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is $1K$ bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

A. 63 B. 64 C. 67 D. 68

gatecse-2007 databases b-tree normal isro2016

Answer key

5.1.16 B Tree: GATE CSE 2008 | Question: 41

A B-tree of order 4 is built from scratch by 10 successive insertions. What is the maximum number of node splitting operations that may take place?

A. 3 B. 4 C. 5 D. 6

gatecse-2008 databases b-tree normal

Answer key

5.1.17 B Tree: GATE CSE 2009 | Question: 44

The following key values are inserted into a B^+ - tree in which order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B^+ - tree is initially empty

$10, 3, 6, 8, 4, 2, 1$

The maximum number of times leaf nodes would get split up as a result of these insertions is

A. 2 B. 3 C. 4 D. 5

gatecse-2009 databases b-tree normal

Answer key

5.1.18 B Tree: GATE CSE 2010 | Question: 18

Consider a B^+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

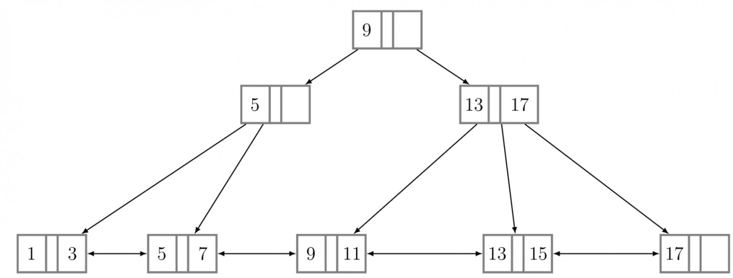
A. 1 B. 2 C. 3 D. 4

gatecse-2010 databases b-tree easy

Answer key

5.1.19 B Tree: GATE CSE 2015 Set 2 | Question: 6

With reference to the B^+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query. "Get all records with a search key greater than or equal to 7 and less than 15 " is ______.



gatecse-2015-set2 databases b-tree normal numerical-answers

Answer key

5.1.20 B Tree: GATE CSE 2015 Set 3 | Question: 46

Consider a B^+ tree in which the search key is 12 bytes long, block size is 1024 bytes, record pointer is 10 bytes long and the block pointer is 8 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is ______.

gatecse-2015-set3 databases b-tree normal numerical-answers

Answer key

5.1.21 B Tree: GATE CSE 2016 Set 2 | Question: 21

B+ Trees are considered BALANCED because.

A. The lengths of the paths from the root to all leaf nodes are all equal.
B. The lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
C. The number of children of any two non-leaf sibling nodes differ by at most 1.
D. The number of records in any two leaf nodes differ by at most 1.

gatecse-2016-set2 databases b-tree normal

Answer key

5.1.22 B Tree: GATE CSE 2017 Set 2 | Question: 49

In a B^+ Tree , if the search-key value is 8 bytes long , the block size is 512 bytes and the pointer size is 2 B , then the maximum order of the B^+ Tree is ____

gatecse-2017-set2 databases b-tree numerical-answers normal

Answer key

5.1.23 B Tree: GATE CSE 2019 | Question: 14

Which one of the following statements is NOT correct about the B^+ tree data structure used for creating an index of a relational database table?

A. B^+ Tree is a height-balanced tree
B. Non-leaf nodes have pointers to data records
C. Key values in each node are kept in sorted order
D. Each leaf node has a pointer to the next leaf node

gatecse-2019 databases b-tree one-mark

Answer key

5.1.24 B Tree: GATE CSE 2024 | Set 1 | Question: 11

In a B^+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

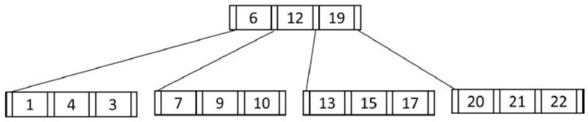
A. Only the root node
B. All leaf nodes
C. All internal nodes
D. Only the leftmost leaf node

gatecse2024-set1 databases b-tree one-mark

Answer key

5.1.25 B Tree: GATE CSE 2025 | Set 1 | Question: 11

Consider the following B^+tree with 5 nodes, in which a node can store at most 3 key values The value 23 is now inserted in the B^+tree. Which of the following options(s) is/are CORRECT?



A. None of the nodes will split.
B. At least one node will split and redistribute.
C. The total number of nodes will remain same.
D. The height of the tree will increase.

gatecse2025-set1 databases b-tree multiple-selects one-mark

Answer key

5.1.26 B Tree: GATE CSE 2025 | Set 2 | Question: 47

In a B^{+}- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes:

$$\mathrm{A} = (49, 77, 83, -), \mathrm{B} = (7, 19, 33, 44), \mathrm{C} = (20^{*}, 22^{*}, 25^{*}, 26^{*})$$

The $*$-marked keys signify that these are data entries in a leaf.

Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent.

A record with key value 23 is inserted into the B^{+}- tree.

The smallest key value in the parent of the leaf that contains 25^{*} is __________. (Answer in integer)

gatecse2025-set2 databases b-tree numerical-answers two-marks

Answer key

5.1.27 B Tree: GATE IT 2004 | Question: 79

Consider a table T in a relational database with a key field K. A B-tree of order p is used as an access structure on K, where p denotes the maximum number of tree pointers in a B-tree index node. Assume that K is $10\ bytes$ long; disk block size is $512\ bytes$; each data pointer P_D is 8 bytes long and each block pointer P_B is 5 bytes long. In order for each B-tree node to fit in a single disk block, the maximum value of p is

A. 20
B. 22
C. 23
D. 32

gateit-2004 databases b-tree normal

Answer key

5.1.28 B Tree: GATE IT 2005 | Question: 23, ISRO2017-67

A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

A. 5
B. 4
C. 3
D. 2

gateit-2005 databases b-tree normal isro2017

Answer key

5.1.29 B Tree: GATE IT 2006 | Question: 61

In a database file structure, the search key field is $9\ bytes$ long, the block size is $512\ bytes$, a record pointer is $7\ bytes$ and a block pointer is $6\ bytes$. The largest possible order of a non-leaf node in a B^{+} tree implementing this file structure is

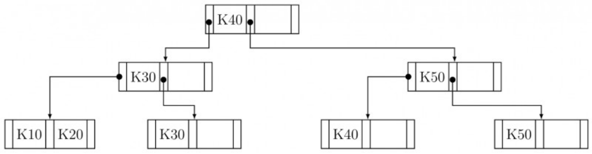
A. 23 B. 24 C. 34 D. 44

gateit-2006 databases b-tree normal

Answer key

5.1.30 B Tree: GATE IT 2007 | Question: 84

Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links.



Keys $K15$ and then $K25$ are inserted into this tree in that order. Exactly how many of the following nodes (disregarding the links) will be present in the tree after the two insertions?

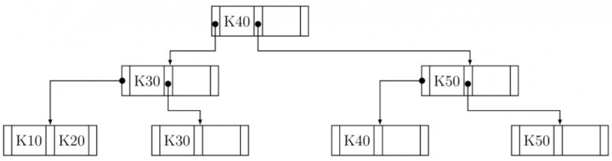
A. 1 B. 2 C. 3 D. 4

gateit-2007 databases b-tree normal

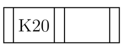
Answer key

5.1.31 B Tree: GATE IT 2007 | Question: 85

Consider the B^+ tree in the adjoining figure, where each node has at most two keys and three links.



Keys $K15$ and then $K25$ are inserted into this tree in that order. Now the key $K50$ is deleted from the B^+ tree resulting after the two insertions made earlier. Consider the following statements about the B^+ tree resulting after this deletion.

i. The height of the tree remains the same.
ii. The node 

(disregarding the links) is present in the tree.
iii. The root node remains unchanged (disregarding the links).

Which one of the following options is true?

A. Statements (i) and (ii) are true
B. Statements (ii) and (iii) are true
C. Statements (iii) and (i) are true
D. All the statements are false

gateit-2007 databases b-tree normal

Answer key

5.2 Candidate Key (5)

5.2.1 Candidate Key: GATE CSE 1994 | Question: 3.7

An instance of a relational scheme $R(A, B, C)$ has distinct values for attribute A. Can you conclude that A is a candidate key for R?

gate1994 databases easy database-normalization candidate-key descriptive

Answer key

5.2.2 Candidate Key: GATE CSE 2011 | Question: 12

Consider a relational table with a single record for each registered student with the following attributes:

1. Registration_Num: Unique registration number for each registered student
2. UID: Unique identity number, unique at the national level for each citizen
3. BankAccount_Num: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attribute stores the primary account number.
4. Name: Name of the student
5. Hostel_Room: Room number of the hostel

Which of the following options is **INCORRECT**?

A. BankAccount_Num is a candidate key
B. Registration_Num can be a primary key
C. UID is a candidate key if all students are from the same country
D. If S is a super key such that $S \cap \text{UID}$ is NULL then $S \cup \text{UID}$ is also a superkey

gatecse-2011 databases normal candidate-key

Answer key

5.2.3 Candidate Key: GATE CSE 2014 Set 2 | Question: 21

The maximum number of superkeys for the relation schema $R(E, F, G, H)$ with E as the key is _____.

gatecse-2014-set2 databases numerical-answers easy candidate-key

Answer key

5.2.4 Candidate Key: GATE CSE 2014 Set 2 | Question: 22

Given an instance of the STUDENTS relation as shown as below

StudentID	StudentName	StudentEmail	StudentAge	CPI
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

For $(\text{StudentName, StudentAge})$ to be a key for this instance, the value X should NOT be equal to______.

gatecse-2014-set2 databases numerical-answers easy candidate-key

Answer key

5.2.5 Candidate Key: GATE CSE 2014 Set 3 | Question: 22

A *prime attribute* of a relation scheme R is an attribute that appears

A. in all candidate keys of R
B. in some candidate key of R
C. in a foreign key of R
D. only in the primary key of R

gatecse-2014-set3 databases easy candidate-key

Answer key

5.3 Conflict Serializable (9)

5.3.1 Conflict Serializable: GATE CSE 2014 Set 1 | Question: 29

Consider the following four schedules due to three transactions (indicated by the subscript) using *read* and *write* on a data item x, denoted by $r(x)$ and $w(x)$ respectively. Which one of them is conflict serializable?

A. $r_1(x); r_2(x); w_1(x); r_3(x); w_2(x);$
B. $r_2(x); r_1(x); w_2(x); r_3(x); w_1(x);$
C. $r_3(x); r_2(x); r_1(x); w_2(x); w_1(x);$
D. $r_2(x); w_2(x); r_3(x); r_1(x); w_1(x);$

gatecse-2014-set1 databases transaction-and-concurrency conflict-serializable normal

Answer key

5.3.2 Conflict Serializable: GATE CSE 2014 Set 2 | Question: 29

Consider the following schedule **S** of transactions $T1, T2, T3, T4$:

T1	**T2**	**T3**	**T4**
	Reads(X)		
		Writes(X)	
		Commit	
Writes(X)			
Commit			
	Writes(Y)		
	Reads(Z)		
	Commit		
			Reads(X)
			Reads(Y)
			Commit

Which one of the following statements is CORRECT?

A. **S** is conflict-serializable but not recoverable
B. **S** is not conflict-serializable but is recoverable
C. **S** is both conflict-serializable and recoverable
D. **S** is neither conflict-serializable not is it recoverable

gatecse-2014-set2 databases transaction-and-concurrency conflict-serializable normal

Answer key

5.3.3 Conflict Serializable: GATE CSE 2014 Set 3 | Question: 29

Consider the transactions $T1, T2,$ and $T3$ and the schedules $S1$ and $S2$ given below.

- $T1 : r1(X); r1(Z); w1(X); w1(Z)$
- $T2 : r2(Y); r2(Z); w2(Z)$
- $T3 : r3(Y); r3(X); w3(Y)$
- $S1 : r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)$
- $S2 : r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)$

Which one of the following statements about the schedules is **TRUE**?

A. Only $S1$ is conflict-serializable.
B. Only $S2$ is conflict-serializable.
C. Both $S1$ and $S2$ are conflict-serializable.
D. Neither $S1$ nor $S2$ is conflict-serializable.

gatecse-2014-set3 databases transaction-and-concurrency conflict-serializable normal

Answer key

5.3.4 Conflict Serializable: GATE CSE 2017 Set 2 | Question: 44

Two transactions T_1 and T_2 are given as

$T_1 : r_1(X)w_1(X)r_1(Y)w_1(Y)$

$T_2 : r_2(Y)w_2(Y)r_2(Z)w_2(Z)$

where $r_i(V)$ denotes a $read$ operation by transaction T_i on a variable V and $w_i(V)$ denotes a $write$ operation by transaction T_i on a variable V. The total number of conflict serializable schedules that can be formed by T_1 and T_2 is ______

gatecse-2017-set2 databases transaction-and-concurrency numerical-answers conflict-serializable

Answer key☟

5.3.5 Conflict Serializable: GATE CSE 2021 Set 1 | Question: 32

Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i. Consider the following two schedules.

- $S_1 : r_1(x)r_1(y)r_2(x)r_2(y)w_2(y)w_1(x)$
- $S_2 : r_1(x)r_2(x)r_2(y)w_2(y)r_1(y)w_1(x)$

Which one of the following options is correct?

A. S_1 is conflict serializable, and S_2 is not conflict serializable
B. S_1 is not conflict serializable, and S_2 is conflict serializable
C. Both S_1 and S_2 are conflict serializable
D. Niether S_1 nor S_2 is conflict serializable

gatecse-2021-set1 databases transaction-and-concurrency conflict-serializable two-marks

Answer key☟

5.3.6 Conflict Serializable: GATE CSE 2021 Set 2 | Question: 32

Let S be the following schedule of operations of three transactions T_1, T_2 and T_3 in a relational database system:

$$R_2(Y), R_1(X), R_3(Z), R_1(Y)W_1(X), R_2(Z), W_2(Y), R_3(X), W_3(Z)$$

Consider the statements P and Q below:

- P: S is conflict-serializable.
- Q: If T_3 commits before T_1 finishes, then S is recoverable.

Which one of the following choices is correct?

A. Both P and Q are true
B. P is true and Q is false
C. P is false and Q is true
D. Both P and Q are false

gatecse-2021-set2 databases transaction-and-concurrency conflict-serializable two-marks

Answer key☟

5.3.7 Conflict Serializable: GATE CSE 2022 | Question: 29

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i, respectively. Consider the schedule S with four transactions.

$S: \ R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$

Which one of the following serial schedules is conflict equivalent to S?

A. $T_1 \to T_3 \to T_4 \to T_2$
B. $T_1 \to T_4 \to T_3 \to T_2$
C. $T_4 \to T_1 \to T_3 \to T_2$
D. $T_3 \to T_1 \to T_4 \to T_2$

gatecse-2022 databases transaction-and-concurrency conflict-serializable two-marks

Answer key☟

5.3.8 Conflict Serializable: GATE CSE 2024 | Set 1 | Question: 36

Consider the following read-write schedule S over three transactions T_1, T_2, and T_3, where the subscripts in the schedule indicate transaction IDs:

$S : r_1(z); w_1(z); r_2(x); r_3(y); w_3(y); r_2(y); w_2(x); w_2(y);$

Which of the following transaction schedules is/are conflict equivalent to S?

A. $T_1T_2T_3$ B. $T_1T_3T_2$ C. $T_3T_2T_1$ D. $T_3T_1T_2$

gatecse2024-set1 databases conflict-serializable multiple-selects two-marks

Answer key☟

5.3.9 Conflict Serializable: GATE CSE 2025 | Set 2 | Question: 43

Consider the database transactions T1 and T2 , and data items X and Y . Which of the schedule(s) is/are conflict serializable?

Transaction T1
R1(X)
W1(Y)
R1(X)
W1(X)
COMMIT(T1)

Transaction T2
W2(X)
W2(Y)
COMMIT(T2)

A. R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)
B. W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
C. R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
D. W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

gatecse2025-set2 databases conflict-serializable multiple-selects two-marks

Answer key☟

5.4 Database Design (1)

5.4.1 Database Design: GATE CSE 1994 | Question: 3.11

State True or False with reason

Logical data independence is easier to achieve than physical data independence.

gate1994 databases normal database-design true-false

Answer key☟

5.5 Database Normalization (56)

5.5.1 Database Normalization: GATE CSE 1987 | Question: 2n

State whether the following statements are TRUE or FALSE:

A relation r with schema (X, Y) satisfies the function dependency $X \rightarrow Y$, The tuples $\langle 1, 2 \rangle$ and $\langle 2, 2 \rangle$ can both be in r simultaneously.

gate1987 databases database-normalization true-false

Answer key☟

5.5.2 Database Normalization: GATE CSE 1988 | Question: 12i

What are the three axioms of functional dependency for the relational databases given by Armstrong.

gate1988 normal descriptive databases database-normalization

Answer key

5.5.3 Database Normalization: GATE CSE 1988 | Question: 12iia

Using Armstrong's axioms of functional dependency derive the following rules:

$$\{x \rightarrow y,\ x \rightarrow z\} \models x \rightarrow yz$$

(Note: $x \rightarrow y$ denotes y is functionally dependent on x, $z \subseteq y$ denotes z is subset of y, and $\models$ means derives).

gate1988 easy descriptive databases database-normalization

Answer key

5.5.4 Database Normalization: GATE CSE 1988 | Question: 12iib

Using Armstrong's axioms of functional dependency derive the following rules:

$$\{x \rightarrow y,\ wy \rightarrow z\} \models xw \rightarrow z$$

(Note: $x \rightarrow y$ denotes y is functionally dependent on x, $z \subseteq y$ denotes z is subset of y, and $\models$ means derives).

gate1988 normal descriptive databases database-normalization

Answer key

5.5.5 Database Normalization: GATE CSE 1988 | Question: 12iic

Using Armstrong's axioms of functional dependency derive the following rules:

$$\{x \rightarrow y,\ z \subset y\} \models x \rightarrow z$$

(Note: $x \rightarrow y$ denotes y is functionally dependent on x, $z \subseteq y$ denotes z is subset of y, and $\models$ means derives).

gate1988 normal descriptive databases database-normalization

Answer key

5.5.6 Database Normalization: GATE CSE 1990 | Question: 2-iv

Match the pairs in the following questions:

(a)	Secondary index	(p)	Function dependency
(b)	Non-procedural query language	(q)	B-tree
(c)	Closure of a set of attributes	(r)	Domain calculus
(d)	Natural join	(s)	Relational algebraic operations

gate1990 match-the-following database-normalization databases

Answer key

5.5.7 Database Normalization: GATE CSE 1990 | Question: 3-ii

Indicate which of the following statements are true:

A relational database which is in 3NF may still have undesirable data redundancy because there may exist:

A. Transitive functional dependencies
B. Non-trivial functional dependencies involving prime attributes on the right-side.
C. Non-trivial functional dependencies involving prime attributes only on the left-side.
D. Non-trivial functional dependencies involving only prime attributes.

gate1990 normal databases database-normalization multiple-selects

Answer key

5.5.8 Database Normalization: GATE CSE 1994 | Question: 3.6

State True or False with reason

There is always a decomposition into Boyce-Codd normal form (BCNF) that is lossless and dependency preserving.

gate1994 databases database-normalization easy true-false

Answer key

5.5.9 Database Normalization: GATE CSE 1995 | Question: 26

Consider the relation scheme $R(A, B, C)$ with the following functional dependencies:

- $A, B \rightarrow C,$
- $C \rightarrow A$

A. Show that the scheme R is in 3NF but not in BCNF.
B. Determine the minimal keys of relation R.

gate1995 databases database-normalization normal descriptive

Answer key

5.5.10 Database Normalization: GATE CSE 1997 | Question: 6.9

For a database relation $R(a, b, c, d)$, where the domains a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold

- $a \rightarrow c$
- $b \rightarrow d$

This relation is

A. in first normal form but not in second normal form
B. in second normal form but not in first normal form
C. in third normal form
D. none of the above

gate1997 databases database-normalization normal

Answer key

5.5.11 Database Normalization: GATE CSE 1998 | Question: 1.34

Which normal form is considered adequate for normal relational database design?

A. $2NF$
B. $5NF$
C. $4NF$
D. $3NF$

gate1998 databases database-normalization easy

Answer key

5.5.12 Database Normalization: GATE CSE 1998 | Question: 26

Consider the following database relations containing the attributes

- Book_id
- Subject_Category_of_book
- Name_of_Author
- Nationality_of_Author

With Book_id as the primary key.

a. What is the highest normal form satisfied by this relation?

b. Suppose the attributes Book_title and Author_address are added to the relation, and the primary key is changed to {Name_of_Author, Book_title}, what will be the highest normal form satisfied by the relation?

gate1998 databases database-normalization normal descriptive

Answer key

5.5.13 Database Normalization: GATE CSE 1999 | Question: 1.24

Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies $C \to F, E \to A, EC \to D, A \to B$. Which one of the following is a key for R?

A. CD
B. EC
C. AE
D. AC

gate1999 databases database-normalization easy

Answer key

5.5.14 Database Normalization: GATE CSE 1999 | Question: 2.7, UGCNET-June2014-III: 25

Consider the schema $R = (S, T, U, V)$ and the dependencies $S \to T, T \to U, U \to V$ and $V \to S$. Let $R = (R1 \text{ and } R2)$ be a decomposition such that $R1 \cap R2 \neq \phi$. The decomposition is

A. not in 2NF
B. in 2NF but not 3NF
C. in 3NF but not in 2NF
D. in both 2NF and 3NF

gate1999 databases database-normalization normal ugcnetjune2014iii

Answer key

5.5.15 Database Normalization: GATE CSE 2000 | Question: 2.24

Given the following relation instance.

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

A. $XY \to Z$ and $Z \to Y$
B. $YZ \to X$ and $Y \to Z$
C. $YZ \to X$ and $X \to Z$
D. $XZ \to Y$ and $Y \to X$

gatecse-2000 databases database-normalization easy

Answer key

5.5.16 Database Normalization: GATE CSE 2001 | Question: 2.23

$R(A, B, C, D)$ is a relation. Which of the following does not have a lossless join, dependency preserving $BCNF$ decomposition?

A. $A \to B, B \to CD$
B. $A \to B, B \to C, C \to D$
C. $AB \to C, C \to AD$
D. $A \to BCD$

gatecse-2001 databases database-normalization normal

Answer key

5.5.17 Database Normalization: GATE CSE 2002 | Question: 1.19

Relation R with an associated set of functional dependencies, F, is decomposed into BCNF. The redundancy (arising out of functional dependencies) in the resulting set of relations is

A. Zero
B. More than zero but less than that of an equivalent 3NF decomposition
C. Proportional to the size of F^+

D. Indeterminate

gatecse-2002 databases database-normalization normal

Answer key

5.5.18 Database Normalization: GATE CSE 2002 | Question: 16

For relation **R = (L, M, N, O, P),** the following dependencies hold:

$M \rightarrow O, NO \rightarrow P, P \rightarrow L$ and $L \rightarrow MN$

R is decomposed into **R1 = (L, M, N, P)** and **R2 = (M, O)**.

A. Is the above decomposition a lossless-join decomposition? Explain.
B. Is the above decomposition dependency-preserving? If not, list all the dependencies that are not preserved.
C. What is the highest normal form satisfied by the above decomposition?

gatecse-2002 databases database-normalization normal descriptive

Answer key

5.5.19 Database Normalization: GATE CSE 2002 | Question: 2.24

Relation R is decomposed using a set of functional dependencies, F, and relation S is decomposed using another set of functional dependencies, G. One decomposition is definitely BCNF, the other is definitely $3NF$, but it is not known which is which. To make a guaranteed identification, which one of the following tests should be used on the decompositions? (Assume that the closures of F and G are available).

A. Dependency-preservation
B. Lossless-join
C. BCNF definition
D. $3NF$ definition

gatecse-2002 databases database-normalization easy

Answer key

5.5.20 Database Normalization: GATE CSE 2002 | Question: 2.25

From the following instance of a relation schema $R(A, B, C)$, we can conclude that:

A	B	C
1	1	1
1	1	0
2	3	2
2	3	2

A. A functionally determines B and B functionally determines C
B. A functionally determines B and B does not functionally determine C
C. B does not functionally determine C
D. A does not functionally determine B and B does not functionally determine C

gatecse-2002 databases database-normalization

Answer key

5.5.21 Database Normalization: GATE CSE 2003 | Question: 85

Consider the following functional dependencies in a database.

Date_of_Birth $\rightarrow$ Age	Age $\rightarrow$ Eligibility
Name $\rightarrow$ Roll_number	Roll_number $\rightarrow$ Name
Course_number $\rightarrow$ Course_name	Course_number $\rightarrow$ Instructor
(Roll_number, Course_number) $\rightarrow$ Grade	

The relation (Roll_number, Name, Date_of_birth, Age) is

A. in second normal form but not in third normal form
B. in third normal form but not in BCNF
C. in BCNF
D. in none of the above

gatecse-2003 databases database-normalization normal

Answer key

5.5.22 Database Normalization: GATE CSE 2004 | Question: 50

The relation scheme $\text{Student Performance (name, courseNo, rollNo, grade)}$ has the following functional dependencies:

- name, courseNo, $\rightarrow$ grade
- rollNo, courseNo $\rightarrow$ grade
- name $\rightarrow$ rollNo
- rollNo $\rightarrow$ name

The highest normal form of this relation scheme is

A. 2NF
B. 3NF
C. BCNF
D. 4NF

gatecse-2004 databases database-normalization normal

Answer key

5.5.23 Database Normalization: GATE CSE 2005 | Question: 29, UGCNET-June2015-III: 9

Which one of the following statements about normal forms is FALSE?

A. BCNF is stricter than 3NF
B. Lossless, dependency-preserving decomposition into 3NF is always possible
C. Lossless, dependency-preserving decomposition into BCNF is always possible
D. Any relation with two attributes is in BCNF

gatecse-2005 databases database-normalization easy ugcnetcse-june2015-paper3

Answer key

5.5.24 Database Normalization: GATE CSE 2005 | Question: 78

Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys R?

A. AE, BE
B. AE, BE, DE
C. AEH, BEH, BCH
D. AEH, BEH, DEH

gatecse-2005 databases database-normalization easy

Answer key

5.5.25 Database Normalization: GATE CSE 2006 | Question: 70

The following functional dependencies are given:

$$AB \rightarrow CD, AF \rightarrow D, DE \rightarrow F, C \rightarrow G, F \rightarrow E, G \rightarrow A$$

Which one of the following options is false?

A. $\{CF\}^* = \{ACDEFG\}$
B. $\{BG\}^* = \{ABCDG\}$
C. $\{AF\}^* = \{ACDEFG\}$
D. $\{AB\}^* = \{ABCDG\}$

gatecse-2006 databases database-normalization normal

Answer key

5.5.26 Database Normalization: GATE CSE 2007 | Question: 62, UGCNET-June2014-II: 47

Which one of the following statements is FALSE?

A. Any relation with two attributes is in BCNF
B. A relation in which every key has only one attribute is in 2NF
C. A prime attribute can be transitively dependent on a key in a 3 NF relation
D. A prime attribute can be transitively dependent on a key in a BCNF relation

gatecse-2007 databases database-normalization normal ugcnetcse-june2014-paper2

Answer key

5.5.27 Database Normalization: GATE CSE 2008 | Question: 69

Consider the following relational schemes for a library database:

Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection(Title, Author, Catalog_no)

with the following functional dependencies:

I. $\text{Title Author} \rightarrow \text{Catalog\_no}$
II. $\text{Catalog\_no} \rightarrow \text{Title Author Publisher Year}$
III. $\text{Publisher Title Year} \rightarrow \text{Price}$

Assume $\{\text{Author, Title}\}$ is the key for both schemes. Which of the following statements is true?

A. Both Book and Collection are in BCNF
B. Both Book and Collection are in 3NF only
C. Book is in 2NF and Collection in 3NF
D. Both Book and Collection are in 2NF only

gatecse-2008 databases database-normalization normal

Answer key

5.5.28 Database Normalization: GATE CSE 2012 | Question: 2

Which of the following is **TRUE**?

A. Every relation in 3NF is also in BCNF
B. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
C. Every relation in BCNF is also in 3NF
D. No relation can be in both BCNF and 3NF

gatecse-2012 databases easy database-normalization

Answer key

5.5.29 Database Normalization: GATE CSE 2013 | Question: 54

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. F= $\{\text{CH}\rightarrow\text{G, A}\rightarrow\text{BC, B}\rightarrow\text{CFH, E}\rightarrow\text{A, F}\rightarrow\text{EG}\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.
How many candidate keys does the relation R have?

A. 3
B. 4
C. 5
D. 6

gatecse-2013 databases database-normalization normal

Answer key

5.5.30 Database Normalization: GATE CSE 2013 | Question: 55

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F = \{\text{CH} \rightarrow \text{G, A} \rightarrow \text{BC, B} \rightarrow \text{CFH, E} \rightarrow \text{A, F} \rightarrow \text{EG}\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

The relation R is

A. in 1NF, but not in 2NF.
B. in 2NF, but not in 3NF.
C. in 3NF, but not in BCNF.
D. in BCNF.

gatecse-2013 databases database-normalization normal

Answer key

5.5.31 Database Normalization: GATE CSE 2014 Set 1 | Question: 21

Consider the relation scheme $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies

$$\{\{E,F\} \to \{G\}, \{F\} \to \{I,J\}, \{E,H\} \to \{K,L\},$$
$$\{K\} \to \{M\}, \{L\} \to \{N\}\}$$

on R. What is the key for R?

A. $\{E,F\}$
B. $\{E,F,H\}$
C. $\{E,F,H,K,L\}$
D. $\{E\}$

gatecse-2014-set1 databases database-normalization normal

Answer key

5.5.32 Database Normalization: GATE CSE 2014 Set 1 | Question: 30

Given the following two statements:

S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.

S2: $AB \to C, D \to E, E \to C$ is a minimal cover for the set of functional dependencies $AB \to C, D \to E, AB \to E, E \to C$.

Which one of the following is **CORRECT**?

A. S1 is TRUE and S2 is FALSE.
B. Both S1 and S2 are TRUE.
C. S1 is FALSE and S2 is TRUE.
D. Both S1 and S2 are FALSE.

gatecse-2014-set1 databases database-normalization normal

Answer key

5.5.33 Database Normalization: GATE CSE 2015 Set 3 | Question: 20

Consider the relation $X(P, Q, R, S, T, U)$ with the following set of functional dependencies

$F = \{$
$\{P, R\} \to \{S, T\},$
$\{P, S, U\} \to \{Q, R\}$
$\}$

Which of the following is the trivial functional dependency in F^+, where F^+ is closure to F?

A. $\{P,R\} \to \{S,T\}$
B. $\{P,R\} \to \{R,T\}$
C. $\{P,S\} \to \{S\}$
D. $\{P,S,U\} \to \{Q\}$

gatecse-2015-set3 databases database-normalization easy

Answer key

5.5.34 Database Normalization: GATE CSE 2016 Set 1 | Question: 21

Which of the following is NOT a superkey in a relational schema with attributes V, W, X, Y, Z and primary key $V\ Y$?

A. $VXYZ$
B. $VWXZ$
C. $VWXY$
D. $VWXYZ$

gatecse-2016-set1 databases database-normalization easy

Answer key

5.5.35 Database Normalization: GATE CSE 2016 Set 1 | Question: 23

A database of research articles in a journal uses the following schema.

$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE})$

The primary key is '$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE})$

and the following functional dependencies exist in the schema.

$$\begin{array}{ll} (\text{VOLUME , NUMBER, STARTPAGE, ENDPAGE}) & \rightarrow \text{TITLE} \\ (\text{VOLUME, NUMBER}) & \rightarrow \text{ YEAR} \\ (\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE}) & \rightarrow \text{PRICE} \end{array}$$

The database is redesigned to use the following schemas

$(\text{VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE})$

$(\text{VOLUME, NUMBER, YEAR})$

Which is the weakest normal form that the new database satisfies, but the old one does not?

A. 1NF
B. 2NF
C. 3NF
D. BCNF

gatecse-2016-set1 databases database-normalization normal

Answer key☟

5.5.36 Database Normalization: GATE CSE 2017 Set 1 | Question: 16

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

- $V \rightarrow W$
- $VW \rightarrow X$
- $Y \rightarrow VX$
- $Y \rightarrow Z$

Which of the following is irreducible equivalent for this set of functional dependencies?

A. $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

B. $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow Z$

C. $V \rightarrow W$
$V \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

D. $V \rightarrow W$
$W \rightarrow X$
$Y \rightarrow V$
$Y \rightarrow X$
$Y \rightarrow Z$

gatecse-2017-set1 databases database-normalization normal

Answer key☟

5.5.37 Database Normalization: GATE CSE 2018 | Question: 42

Consider the following four relational schemas. For each schema , all non-trivial functional dependencies are listed, The **bolded** attributes are the respective primary keys.

Schema I: Registration(rollno, courses)

Field ‘courses’ is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency

rollno $\rightarrow$ courses

Schema II: Registration (rollno, coursid, email)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ email

email $\rightarrow$ rollno

Schema III: Registration (rollno, courseid, marks, grade)

Non-trivial functional dependencies:

rollno, courseid, $\rightarrow$ marks, grade marks $\rightarrow$ grade

Schema IV: Registration (rollno, courseid, credit)

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ credit

courseid $\rightarrow$ credit

Which one of the relational schemas above is in 3NF but not in BCNF?

A. Schema I
B. Schema II
C. Schema III
D. Schema IV

gatecse-2018 databases database-normalization normal two-marks

Answer key

5.5.38 Database Normalization: GATE CSE 2019 | Question: 32

Let the set of functional dependencies $F = \{QR \rightarrow S,\ R \rightarrow P,\ S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where $Y = (PR)$ and $Z = (QRS)$.

Consider the two statements given below.

I. Both Y and Z are in BCNF
II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

A. Both I and II
B. I only
C. II only
D. Neither I nor II

gatecse-2019 databases database-normalization two-marks

Answer key

5.5.39 Database Normalization: GATE CSE 2020 | Question: 36

Consider a relational table R that is in $3NF$, but not in BCNF. Which one of the following statements is TRUE?

A. R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a prime attribute.
B. R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.
C. R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key
D. A cell in R holds a set instead of an atomic value.

gatecse-2020 databases database-normalization two-marks

Answer key

5.5.40 Database Normalization: GATE CSE 2021 Set 1 | Question: 33

Consider the relation $R(P,Q,S,T,X,Y,Z,W)$ with the following functional dependencies.

$$PQ \rightarrow X; \quad P \rightarrow YX; \quad Q \rightarrow Y; \quad Y \rightarrow ZW$$

Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.

- $D_1: \quad R = [(P,Q,S,T);\ (P,T,X);\ (Q,Y);\ (Y,Z,W)]$
- $D_2: \quad R = [(P,Q,S);\ (T,X);\ (Q,Y);\ (Y,Z,W)]$

Which one of the following options is correct?

A. D_1 is a lossless decomposition, but D_2 is a lossy decomposition
B. D_1 is a lossy decomposition, but D_2 is a lossless decomposition
C. Both D_1 and D_2 are lossless decompositions

D. Both D_1 and D_2 are lossy decompositions

gatecse-2021-set1 databases database-normalization two-marks

Answer key

5.5.41 Database Normalization: GATE CSE 2021 Set 2 | Question: 40

Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S, and T:

- $P \to QR$
- $RS \to T$

Which of the following functional dependencies can be inferred from the above functional dependencies?

A. $PS \to T$ B. $R \to T$ C. $P \to R$ D. $PS \to Q$

gatecse-2021-set2 multiple-selects databases database-normalization two-marks

Answer key

5.5.42 Database Normalization: GATE CSE 2022 | Question: 21

Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies.

$AB \to C;\ BC \to D;\ C \to E;$

The number of superkeys in the relation R is ______________ .

gatecse-2022 numerical-answers databases database-normalization one-mark

Answer key

5.5.43 Database Normalization: GATE CSE 2022 | Question: 4

In a relational data model, which one of the following statements is TRUE?

A. A relation with only two attributes is always in BCNF.
B. If all attributes of a relation are prime attributes, then the relation is in BCNF.
C. Every relation has at least one non-prime attribute.
D. BCNF decompositions preserve functional dependencies.

gatecse-2022 databases database-normalization one-mark

Answer key

5.5.44 Database Normalization: GATE CSE 2024 | Set 1 | Question: 12

Which of the following statements about a relation $\mathbf{R}$ in first normal form (1NF) is/are TRUE?

A. $\mathbf{R}$ can have a multi-attribute key
B. $\mathbf{R}$ cannot have a foreign key
C. $\mathbf{R}$ cannot have a composite attribute
D. $\mathbf{R}$ cannot have more than one candidate key

gatecse2024-set1 multiple-selects databases database-normalization one-mark

Answer key

5.5.45 Database Normalization: GATE CSE 2024 | Set 1 | Question: 34

The symbol $\to$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

A. $(X, Y) \to (Z, W)$ implies $X \to (Z, W)$
B. $(X, Y) \to (Z, W)$ implies $(X, Y) \to Z$

C. $((X, Y) \rightarrow Z$ and $W \rightarrow Y)$ implies $(X, W) \rightarrow Z$
D. $(X \rightarrow Y$ and $Y \rightarrow Z)$ implies $X \rightarrow Z$

gatecse2024-set1 multiple-selects databases database-normalization two-marks

Answer key☟

5.5.46 Database Normalization: GATE CSE 2024 | Set 2 | Question: 46

A functional dependency $F : X \rightarrow Y$ is termed as a useful functional dependency if and only if it satisfies all the following three conditions:

- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is __________.

gatecse2024-set2 numerical-answers databases database-normalization two-marks

Answer key☟

5.5.47 Database Normalization: GATE CSE 2025 | Set 2 | Question: 36

Consider the following relational schema along with all the functional dependencies that hold on them.

$$R1(A, B, C, D, E) : \{D \rightarrow E, EA \rightarrow B, EB \rightarrow C\}$$
$$R2(A, B, C, D) : \{A \rightarrow D, A \rightarrow B, C \rightarrow A\}$$

Which of the following statement(s) is/are TRUE?

A. $R1$ is in 3 NF
B. $R2$ is in 3 NF
C. $R1$ is NOT in 3 NF
D. $R2$ is NOT in 3 NF

gatecse2025-set2 databases database-normalization multiple-selects two-marks

Answer key☟

5.5.48 Database Normalization: GATE DS&AI 2024 | Question: 36

Given the relational schema $R = (U, V, W, X, Y, Z)$ and the set of functional dependencies:

$$\{U \rightarrow V, U \rightarrow W, WX \rightarrow Y, WX \rightarrow Z, V \rightarrow X\}$$

Which of the following functional dependencies can be derived from the above set?

A. $VW \rightarrow YZ$
B. $WX \rightarrow YZ$
C. $VW \rightarrow U$
D. $VW \rightarrow Y$

gate-ds-ai-2024 databases database-normalization multiple-selects two-marks

Answer key☟

5.5.49 Database Normalization: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 26

Given the following relation instances

X	Y	Z
1	4	2
1	5	3
1	4	3
1	5	2
3	2	1

Which of the following conditions is/are **TRUE**?

A. $\mathrm{XY}->\mathrm{Z}$ and $\mathrm{Z}->\mathrm{Y}$
B. $\mathrm{YZ}->\mathrm{X}$ and $\mathrm{X}->>\mathrm{Y}$
C. $\mathrm{Y}->\mathrm{X}$ and $\mathrm{Y}->>\mathrm{X}$
D. $\mathrm{XZ}->\mathrm{Y}$ and $\mathrm{Y}->\mathrm{X}$

gateda-sample-paper-2024 database-normalization

Answer key

5.5.50 Database Normalization: GATE IT 2004 | Question: 75

A relation Empdtl is defined with attributes empcode (unique), name, street, city, state and pincode. For any pincode, there is only one city and state. Also, for any given street, city and state, there is just one pincode. In normalization terms, Empdtl is a relation in

A. 1NF only
B. 2NF and hence also in 1NF
C. 3NF and hence also in 2NF and 1NF
D. BCNF and hence also in 3NF, 2NF and 1NF

gateit-2004 databases database-normalization normal

Answer key

5.5.51 Database Normalization: GATE IT 2005 | Question: 22

A table has fields F_1, F_2, F_3, F_4, F_5 with the following functional dependencies

- $F_1 \rightarrow F_3, F_2 \rightarrow F_4, (F_1 . F_2) \rightarrow F_5$

In terms of Normalization, this table is in

A. 1 NF
B. 2 NF
C. 3 NF
D. None of these

gateit-2005 databases database-normalization easy

Answer key

5.5.52 Database Normalization: GATE IT 2005 | Question: 70

In a schema with attributes A, B, C, D and E following set of functional dependencies are given

- $A \rightarrow B$
- $A \rightarrow C$
- $CD \rightarrow E$
- $B \rightarrow D$
- $E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

A. $CD \rightarrow AC$
B. $BD \rightarrow CD$
C. $BC \rightarrow CD$
D. $AC \rightarrow BC$

gateit-2005 databases database-normalization normal

Answer key

5.5.53 Database Normalization: GATE IT 2006 | Question: 60

Consider a relation R with five attributes $V, W, X, Y,$ and $Z.$ The following functional dependencies hold: $VY \to W, WX \to Z,$ and $ZY \to V.$

Which of the following is a candidate key for R?

A. VXZ B. VXY C. $VWXY$ D. $VWXYZ$

gateit-2006 databases database-normalization normal

Answer key

5.5.54 Database Normalization: GATE IT 2008 | Question: 61

Let $R(A, B, C, D)$ be a relational schema with the following functional dependencies : $A \to B$, $B \to C$, $C \to D$ and $D \to B$. The decomposition of R into $(A, B), (B, C), (B, D)$

A. gives a lossless join, and is dependency preserving
B. gives a lossless join, but is not dependency preserving
C. does not give a lossless join, but is dependency preserving
D. does not give a lossless join and is not dependency preserving

gateit-2008 databases database-normalization normal

Answer key

5.5.55 Database Normalization: GATE IT 2008 | Question: 62

Let $R(A, B, C, D, E, P, G)$ be a relational schema in which the following functional dependencies are known to hold: $AB \to CD, DE \to P, C \to E, P \to C$ and $B \to G.$ The relational schema R is

A. in BCNF
B. in 3NF, but not in BCNF
C. in 2NF, but not in 3NF
D. not in 2NF

gateit-2008 databases database-normalization normal

Answer key

5.5.56 Database Normalization: GATE2001-1.23, UGCNET-June2012-III: 18

Consider a schema $R(A, B, C, D)$ and functional dependencies $A \to B$ and $C \to D.$ Then the decomposition of R into $R_1(A, B)$ and $R_2(C, D)$ is

A. dependency preserving and lossless join
B. lossless join but not dependency preserving
C. dependency preserving but not lossless join
D. not dependency preserving and not lossless join

gate1998 databases ugcnetcse-june2012-paper3 database-normalization

Answer key

5.6 Decomposition (1)

5.6.1 Decomposition: GATE DA 2025 | Question: 6

If a relational decomposition is not dependency-preserving, which one of the following relational operators will be executed more frequently in order to maintain the dependencies?

A. Selection B. Projection C. Join D. Set union

gateda-2025 databases decomposition relational-algebra easy one-mark

Answer key

5.7 ER Diagram (12)

5.7.1 ER Diagram: GATE CSE 2005 | Question: 75

Let E_1 and E_2 be two entities in an E/R diagram with simple-valued attributes. R_1 and R_2 are two relationships between E_1 and E_2, where R_1 is one-to-many and R_2 is many-to-many. R_1 and R_2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

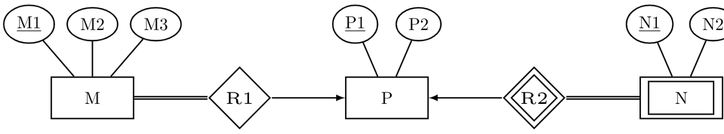
A. 2
B. 3
C. 4
D. 5

gatecse-2005 databases er-diagram normal

Answer key

5.7.2 ER Diagram: GATE CSE 2008 | Question: 82

Consider the following ER diagram



The minimum number of tables needed to represent $M, N, P, R1, R2$ is

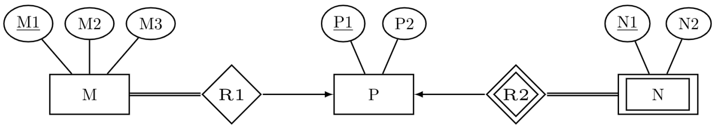
A. 2
B. 3
C. 4
D. 5

gatecse-2008 databases er-diagram normal

Answer key

5.7.3 ER Diagram: GATE CSE 2008 | Question: 83

Consider the following ER diagram



The minimum number of tables needed to represent $M, N, P, R1, R2$ is

Which of the following is a correct attribute set for one of the tables for the minimum number of tables needed to represent $M, N, P, R1, R2$?

A. $M1, M2, M3, P1$
B. $M1, P1, N1, N2$
C. $M1, P1, N1$
D. $M1, P1$

gatecse-2008 databases er-diagram normal

Answer key

5.7.4 ER Diagram: GATE CSE 2012 | Question: 14

Given the basic ER and relational models, which of the following is **INCORRECT**?

A. An attribute of an entity can have more than one value
B. An attribute of an entity can be composite
C. In a row of a relational table, an attribute can have more than one value
D. In a row of a relational table, an attribute can have exactly one value or a NULL value

gatecse-2012 databases normal er-diagram

Answer key

5.7.5 ER Diagram: GATE CSE 2015 Set 1 | Question: 41

Consider an Entity-Relationship (ER) model in which entity sets E_1 and E_2 are connected by an $m:n$ relationship R_{12}. E_1 and E_3 are connected by a $1:n$ (1 on the side of E_1 and n on the side of E_3) relationship R_{13}.

E_1 has two-singled attributes a_{11} and a_{12} of which a_{11} is the key attribute. E_2 has two singled-valued attributes a_{21} and a_{22} of which a_{21} is the key attribute. E_3 has two single-valued attributes a_{31} and a_{32} of which a_{31} is the key attribute. The relationships do not have any attributes.

If a relational model is derived from the above ER model, then the minimum number of relations that would be generated if all relation are in 3NF is________________.

gatecse-2015-set1 databases er-diagram normal numerical-answers

Answer key☟

5.7.6 ER Diagram: GATE CSE 2017 Set 2 | Question: 17

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

A. Relationship R is one-to-many and the participation of A in R is total
B. Relationship R is one-to-many and the participation of A in R is partial
C. Relationship R is many-to-one and the participation of A in R is total
D. Relationship R is many-to-one and the participation of A in R is partial

gatecse-2017-set2 databases er-diagram normal

Answer key☟

5.7.7 ER Diagram: GATE CSE 2018 | Question: 11

In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in R and that the cardinality of E1 is greater than the cardinality of E2.

Which one of the following is true about R?

A. Every entity in E1 is associated with exactly one entity in E2
B. Some entity in E1 is associated with more than one entity in E2
C. Every entity in E2 is associated with exactly one entity in E1
D. Every entity in E2 is associated with at most one entity in E1

gatecse-2018 databases er-diagram normal one-mark

Answer key☟

5.7.8 ER Diagram: GATE CSE 2020 | Question: 14

Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

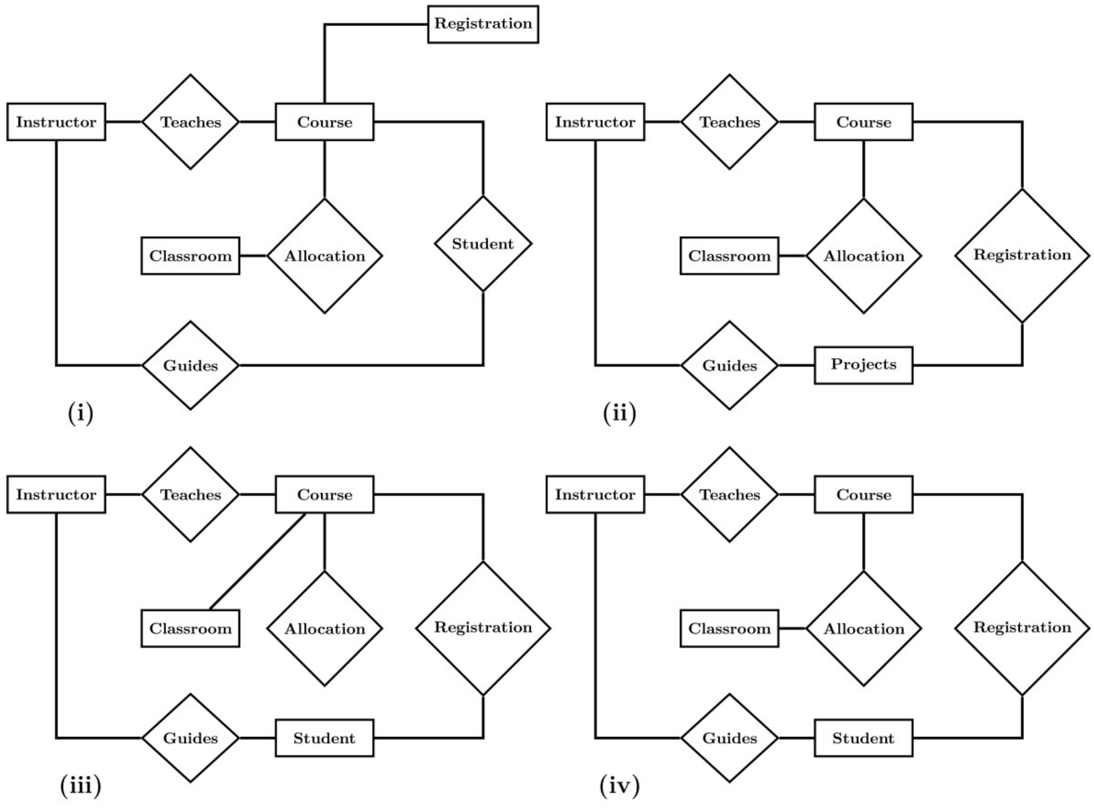
A. Diamonds with double/bold border
B. Rectangles with double/bold border
C. Ovals with double/bold border
D. Ovals that contain underlined identifiers

gatecse-2020 databases er-diagram one-mark

Answer key☟

5.7.9 ER Diagram: GATE CSE 2024 | Set 1 | Question: 10

Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



A. (i)
B. (ii)
C. (iii)
D. (iv)

gatecse2024-set1 databases er-diagram one-mark

Answer key☟

5.7.10 ER Diagram: GATE CSE 2024 | Set 2 | Question: 10

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

A. The weak entity set MUST have total participation in the identifying relationship
B. The owner entity set MUST have total participation in the identifying relationship
C. Both weak and owner entity sets MUST have total participation in the identifying relationship
D. Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

gatecse2024-set2 databases er-diagram one-mark

Answer key☟

5.7.11 ER Diagram: GATE IT 2004 | Question: 73

Consider the following entity relationship diagram (ERD), where two entities $E1$ and $E2$ have a relation R of cardinality 1:m.

The attributes of $E1$ are $A11$, $A12$ and $A13$ where $A11$ is the key attribute. The attributes of $E2$ are $A21$, $A22$ and $A23$ where $A21$ is the key attribute and $A23$ is a multi-valued attribute. Relation R does not have any attribute. A relational database containing minimum number of tables with each table satisfying the requirements of the third normal form ($3NF$) is designed from the above ERD. The number of tables in the database is

A. 2
B. 3
C. 5
D. 4

gateit-2004 databases er-diagram normal

Answer key

5.7.12 ER Diagram: GATE IT 2005 | Question: 21

Consider the entities 'hotel room', and 'person' with a many to many relationship 'lodging' as shown below:

If we wish to store information about the rent payment to be made by person (s) occupying different hotel rooms, then this information should appear as an attribute of

A. Person
B. Hotel Room
C. Lodging
D. None of these

gateit-2005 databases er-diagram easy

Answer key

5.8 Functional Dependency (2)

5.8.1 Functional Dependency: GATE CSE 2025 | Set 1 | Question: 37

Consider a relational schema team (name,city,owner), with functional dependencies $\{\text{name} \rightarrow \text{city}, \text{name} \rightarrow \text{owner}\}$.

The relation team is decomposed into two relations, $\text{t1}(\text{name,city})$ and $\text{t2}(\text{name, owner})$. Which of the following statement(s) is/are TRUE?

A. The relation team is NOT in BCNF .
B. The relations $t1$ and $t2$ are in BCNF.
C. The decomposition constitutes a lossless join.
D. The relation team is NOT in 3 NF .

gatecse2025-set1 databases functional-dependency database-normalization multiple-selects two-marks

Answer key

5.8.2 Functional Dependency: GATE DA 2025 | Question: 47

Consider a database relation R with attributes ABCDEFG, and having the following functional dependencies:

$$\text{A} \rightarrow \text{BCEF} \quad \text{E} \rightarrow \text{DG} \quad \text{BC} \rightarrow \text{A}$$

Which of the following statements is/are correct?

A. A is the only candidate key of R
B. A, BC are the candidate keys of R
C. $\text{A}, \text{BC}, \text{E}$ are the candidate keys of R
D. Relation R is not in Boyce-Codd Normal Form (BCNF)

gateda-2025 databases functional-dependency multiple-selects two-marks

Answer key

5.9 Indexing (13)

5.9.1 Indexing: GATE CSE 1989 | Question: 4-xiv

For secondary key processing which of the following file organizations is preferred? Give a one line justification:

A. Indexed sequential file organization.
B. Two-way linked list.
C. Inverted file organization.
D. Sequential file organization.

gate1989 normal databases indexing descriptive

Answer key

5.9.2 Indexing: GATE CSE 1990 | Question: 10b

One giga bytes of data are to be organized as an indexed-sequential file with a uniform blocking factor 8. Assuming a block size of 1 Kilo bytes and a block refrencing pointer size of 32 bits, find out the number of levels of indexing that would be required and the size of the index at each level. Determine also the size of the master index. The referencing capability (fanout ratio) per block of index storage may be considered to be 32.

gate1990 databases indexing descriptive

Answer key

5.9.3 Indexing: GATE CSE 1993 | Question: 14

An ISAM (indexed sequential) file consists of records of size 64 bytes each, including key field of size 14 bytes. An address of a disk block takes 2 bytes. If the disk block size is 512 bytes and there are $16K$ records, compute the size of the data and index areas in terms of number blocks. How many levels of tree do you have for the index?

gate1993 databases indexing normal descriptive

Answer key

5.9.4 Indexing: GATE CSE 1998 | Question: 1.35

There are five records in a database.

Name	**Age**	**Occupation**	**Category**
Rama	27	CON	A
Abdul	22	ENG	A
Jennifer	28	DOC	B
Maya	32	SER	D
Dev	24	MUS	C

There is an index file associated with this and it contains the values $1, 3, 2, 5$ and 4. Which one of the fields is the index built from?

A. Age
B. Name
C. Occupation
D. Category

gate1998 databases indexing normal

Answer key

5.9.5 Indexing: GATE CSE 2008 | Question: 16, ISRO2016-60

A clustering index is defined on the fields which are of type

A. non-key and ordering
B. non-key and non-ordering
C. key and ordering
D. key and non-ordering

gatecse-2008 easy databases indexing isro2016

Answer key

5.9.6 Indexing: GATE CSE 2008 | Question: 70

Consider a file of 16384 records. Each record is 32 bytes long and its key field is of size 6 bytes. The file is ordered on a non-key field, and the file organization is unspanned. The file is stored in a file system with block size 1024 bytes, and the size of a block pointer is 10 bytes. If the secondary index is built on the key field of the file, and a multi-level index scheme is used to store the secondary index, the number of first-level and second-level blocks in the multi-level index are respectively

A. 8 and 0
B. 128 and 6
C. 256 and 4
D. 512 and 5

gatecse-2008 databases indexing normal

Answer key

5.9.7 Indexing: GATE CSE 2011 | Question: 39

Consider a relational table r with sufficient number of records, having attributes $A_1, A_2, \ldots, A_n$ and let $1 \le p \le n$. Two queries $Q1$ and $Q2$ are given below.

- $Q1 : \pi_{A_1,\ldots,A_p}\left(\sigma_{A_p=c}(r)\right)$ where c is a constant
- $Q2 : \pi_{A_1,\ldots,A_p}\left(\sigma_{c_1 \le A_p \le c_2}(r)\right)$ where c_1 and c_2 are constants.

The database can be configured to do ordered indexing on A_p or hashing on A_p. Which of the following statements is **TRUE**?

A. Ordered indexing will always outperform hashing for both queries
B. Hashing will always outperform ordered indexing for both queries
C. Hashing will outperform ordered indexing on $Q1$, but not on $Q2$
D. Hashing will outperform ordered indexing on $Q2$, but not on $Q1$

gatecse-2011 databases indexing normal

Answer key

5.9.8 Indexing: GATE CSE 2013 | Question: 15

An index is clustered, if

A. it is on a set of fields that form a candidate key
B. it is on a set of fields that include the primary key
C. the data records of the file are organized in the same order as the data entries of the index
D. the data records of the file are organized not in the same order as the data entries of the index

gatecse-2013 databases indexing normal

Answer key

5.9.9 Indexing: GATE CSE 2015 Set 1 | Question: 24

A file is organized so that the ordering of the data records is the same as or close to the ordering of data entries in some index. Then that index is called

A. Dense B. Sparse C. Clustered D. Unclustered

gatecse-2015-set1 databases indexing easy

Answer key

5.9.10 Indexing: GATE CSE 2020 | Question: 54

Consider a database implemented using B^+ tree for file indexing and installed on a disk drive with block size of $4\,\text{KB}$. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B^+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _______

gatecse-2020 numerical-answers databases b-tree indexing two-marks

Answer key

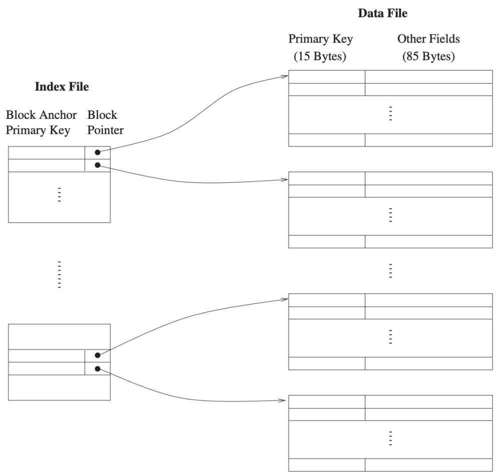
5.9.11 Indexing: GATE CSE 2021 Set 2 | Question: 21

A data file consisting of $1,50,000$ student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called ANum of size 12 bytes. Suppose an index file with records consisting of two fields, ANum value and the record pointer the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is ________ gatecse-2021-set2 numerical-answers databases indexing one-mark

Answer key☟

5.9.12 Indexing: GATE CSE 2023 | Question: 52

Consider a database of fixed-length records, stored as an ordered file. The database has $25,000$ records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____________.

gatecse-2023 databases file-system indexing numerical-answers two-marks

Answer key☟

5.9.13 Indexing: GATE CSE 2024 | Set 2 | Question: 16

Which of the following file organizations is/are $\mathrm{I/O}$ efficient for the scan operation in DBMS?

A. Sorted
B. Heap
C. Unclustered tree index
D. Unclustered hash index

gatecse2024-set2 databases multiple-selects indexing one-mark

Answer key☟

5.10 Joins (7)

5.10.1 Joins: GATE CSE 2004 | Question: 14

Consider the following relation schema pertaining to a students database:

- Students (<u>rollno,</u> name, address)
- Enroll (<u>rollno, courseno</u>, coursename)

where the primary keys are shown underlined. The number of tuples in the student and Enroll tables are 120 and 8 respectively. What are the maximum and minimum number of tuples that can be present in (Student * Enroll), where '*' denotes natural join?

A. $8, 8$
B. $120, 8$
C. $960, 8$
D. $960, 120$

gatecse-2004 databases easy joins natural-join

Answer key☟

5.10.2 Joins: GATE CSE 2012 | Question: 50

Consider the following relations A, B and C :

A

ID	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

ID	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

ID	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following relational algebra expression contain? Assume that the schema of $A \cup B$ is the same as that of A.

$$(A \cup B) \bowtie_{A.Id>40 \vee C.Id<15} C$$

A. 7
B. 4
C. 5
D. 9

gatecse-2012 databases joins normal

Answer key

5.10.3 Joins: GATE CSE 2014 Set 2 | Question: 30

Consider a join (relation algebra) between relations $r(R)$ and $s(S)$ using the nested loop method. There are 3 buffers each of size equal to disk block size, out of which one buffer is reserved for intermediate results. Assuming $\text{size}(r(R)) < \text{size}(s(S))$, the join will have fewer number of disk block accesses if

A. relation $r(R)$ is in the outer loop.
B. relation $s(S)$ is in the outer loop.
C. join selection factor between $r(R)$ and $s(S)$ is more than 0.5.
D. join selection factor between $r(R)$ and $s(S)$ is less than 0.5.

gatecse-2014-set2 databases normal joins

Answer key

5.10.4 Joins: GATE IT 2005 | Question: 82a

A database table T_1 has 2000 records and occupies 80 disk blocks. Another table T_2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T_1 and one block of records for T_2 simultaneously at any point in time. No index is available on either table.

If Nested-loop join algorithm is employed to perform the join, with the most appropriate choice of table to be used in outer loop, the number of block accesses required for reading the data are

A. 800000
B. 40080
C. 32020
D. 100

gateit-2005 databases normal joins

Answer key

5.10.5 Joins: GATE IT 2005 | Question: 82b

A database table T_1 has 2000 records and occupies 80 disk blocks. Another table T_2 has 400 records and occupies 20 disk blocks. These two tables have to be joined as per a specified join condition that needs to be evaluated for every pair of records from these two tables. The memory buffer space available can hold exactly one block of records for T_1 and one block of records for T_2 simultaneously at any point in time. No index is available on either table.

If, instead of Nested-loop join, Block nested-loop join is used, again with the most appropriate choice of table in the outer loop, the reduction in number of block accesses required for reading the data will be

A. 0 B. 30400 C. 38400 D. 798400

gateit-2005 databases normal joins

Answer key

5.10.6 Joins: GATE IT 2006 | Question: 14

Consider the relations $r_1(\mathrm{P, Q, R})$ and $r_2(\mathrm{R, S, T})$ with primary keys P and R respectively. The relation r_1 contains 2000 tuples and r_2 contains 2500 tuples. The maximum size of the join $r_1 \bowtie r_2$ is :

A. 2000 B. 2500 C. 4500 D. 5000

gateit-2006 databases joins natural-join normal

Answer key

5.10.7 Joins: GATE IT 2007 | Question: 68

Consider the following relation schemas :

- b-Schema = (b-name, b-city, assets)
- a-Schema = (a-num, b-name, bal)
- d-Schema = (c-name, a-number)

Let branch, account and depositor be respectively instances of the above schemas. Assume that account and depositor relations are much bigger than the branch relation.

Consider the following query:

$\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"} \wedge \text{ bal < 0}}$ (branch ⋈ (account ⋈ depositor)

Which one of the following queries is the most efficient version of the above query ?

A. $\Pi_{\text{c-name}}$ ($\sigma_{\text{bal < 0}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ account) ⋈ depositor)
B. $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ ($\sigma_{\text{bal < 0}}$ account ⋈ depositor))
C. $\Pi_{\text{c-name}}$ (($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ $\sigma_{\text{b-city = "Agra"} \wedge \text{ bal < 0}}$ account) ⋈ depositor)
D. $\Pi_{\text{c-name}}$ ($\sigma_{\text{b-city = "Agra"}}$ branch ⋈ ($\sigma_{\text{b-city = "Agra"} \wedge \text{ bal < 0}}$ account ⋈ depositor))

gateit-2007 databases joins relational-algebra normal

Answer key

5.11 Multivalued Dependency 4nf (1)

5.11.1 Multivalued Dependency 4nf: GATE IT 2007 | Question: 67

Consider the following implications relating to functional and multivalued dependencies given below, which may or may not be correct.

i. if $A \rightarrow\rightarrow B$ and $A \rightarrow\rightarrow C$ then $A \rightarrow BC$
ii. if $A \rightarrow B$ and $A \rightarrow C$ then $A \rightarrow\rightarrow BC$
iii. if $A \rightarrow\rightarrow BC$ and $A \rightarrow B$ then $A \rightarrow C$
iv. if $A \rightarrow BC$ and $A \rightarrow B$ then $A \rightarrow\rightarrow C$

Exactly how many of the above implications are valid?

A. 0 B. 1 C. 2 D. 3

gateit-2007 databases database-normalization multivalued-dependency-4nf normal

Answer key

5.12 Natural Join (3)

5.12.1 Natural Join: GATE CSE 2005 | Question: 30

Let r be a relation instance with schema R = (A, B, C, D). We define $r_1 = \pi_{A,B,C}(R)$ and $r_2 = \pi_{A,D}(r)$. Let $s = r_1 * r_2$ where $*$ denotes natural join. Given that the decomposition of r into r_1 and r_2 is lossy, which one of the following is TRUE?

A. $s \subset r$ B. $r \cup s = r$ C. $r \subset s$ D. $r * s = s$

gatecse-2005 databases relational-algebra natural-join normal

Answer key

5.12.2 Natural Join: GATE CSE 2010 | Question: 43

The following functional dependencies hold for relations $R(A, B, C)$ and $S(B, D, E)$.

- $B \rightarrow A$
- $A \rightarrow C$

The relation R contains 200 tuples and the relation S contains 100 tuples. What is the maximum number of tuples possible in the natural join $R \bowtie S$?

A. 100 B. 200 C. 300 D. 2000

gatecse-2010 databases normal natural-join database-normalization

Answer key

5.12.3 Natural Join: GATE CSE 2015 Set 2 | Question: 32

Consider two relations $R_1(A, B)$ with the tuples $(1, 5), (3, 7)$ and $R_2(A, C) = (1, 7), (4, 9)$.

Assume that $R(A, B, C)$ is the full natural outer join of R_1 and R_2. Consider the following tuples of the form (A, B, C) :

$a = (1, 5, null), b = (1, null, 7), c = (3, null, 9), d = (4, 7, null), e = (1, 5, 7),$
$f = (3, 7, null), g = (4, null, 9).$

Which one of the following statements is correct?

A. R contains a, b, e, f, g but not c, d.
B. R contains all a, b, c, d, e, f, g.
C. R contains e, f, g but not a, b.
D. R contains e but not f, g.

gatecse-2015-set2 databases normal natural-join

Answer key

5.13 Query (1)

5.13.1 Query: GATE CSE 1997 | Question: 76-b

Consider the following relational database schema:

- EMP (eno name, age)
- PROJ (pno name)
- INVOLVED (eno, pno)

EMP contains information about employees. PROJ about projects and involved about which employees involved in which projects. The underlined attributes are the primary keys for the respective relations.

State in English (in not more than 15 words)

What the following relational algebra expressions are designed to determine

i. $\Pi_{eno}(\text{INVOLVED}) - \Pi_{eno}((\Pi_{eno}(\text{INVOLVED}) \times \Pi_{pno}(\text{PROJ})) - \text{INVOLVED})$
ii. $\Pi_{age}(\text{EMP}) - \Pi_{age}(\sigma_{E.age<Emp.age}((\rho E(\text{EMP}) \times \text{EMP}))$

(Note: $\rho E(\text{EMP})$ conceptually makes a copy of EMP and names it E (ρ is called the rename operator))

gate1997 databases sql descriptive normal relational-algebra query

Answer key

5.14 Referential Integrity (5)

5.14.1 Referential Integrity: GATE CSE 1997 | Question: 6.10, ISRO2016-54

Let $R(a,b,c)$ and $S(d,e,f)$ be two relations in which d is the foreign key of S that refers to the primary key of R. Consider the following four operations R and S

I. Insert into R
II. Insert into S
III. Delete from R
IV. Delete from S

Which of the following can cause violation of the referential integrity constraint above?

A. Both I and IV
B. Both II and III
C. All of these
D. None of these

gate1997 databases referential-integrity easy isro2016

Answer key

5.14.2 Referential Integrity: GATE CSE 2005 | Question: 76

The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on-delete cascade.

A	C
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(2,4)$ is deleted is:

A. $(3,4)$ and $(6,4)$
B. $(5,2)$ and $(7,2)$
C. $(5,2),(7,2)$ and $(9,5)$
D. $(3,4),(4,3)$ and $(6,4)$

gatecse-2005 databases referential-integrity normal

Answer key

5.14.3 Referential Integrity: GATE CSE 2012 | Question: 43

Suppose $R_1(\underline{A},B)$ and $R_2(\underline{C},D)$ are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2. If data in r_1 and r_2 satisfy referential integrity constraints, which of the following is **ALWAYS TRUE**?

A. $\prod_B(r_1) - \prod_C(r_2) = \varnothing$
B. $\prod_C(r_2) - \prod_B(r_1) = \varnothing$
C. $\prod_B(r_1) = \prod_C(r_2)$
D. $\prod_B(r_1) - \prod_C(r_2) \neq \varnothing$

gatecse-2012 databases relational-algebra normal referential-integrity

Answer key

5.14.4 Referential Integrity: GATE CSE 2017 Set 2 | Question: 19

Consider the following tables $T1$ and $T2$.

$T1$

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

$T2$

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table $T1$ **P** is the primary key and **Q** is the foreign key referencing **R** in table $T2$ with on-delete cascade and on-update cascade. In table $T2,$ **R** is the primary key and **S** is the foreign key referencing **P** in table $T1$ with on-delete set NULL and on-update cascade. In order to delete record $\langle 3, 8 \rangle$ from the table $T1,$ the number of additional records that need to be deleted from table $T1$ is _______

gatecse-2017-set2 databases numerical-answers referential-integrity normal

Answer key

5.14.5 Referential Integrity: GATE CSE 2021 Set 2 | Question: 6

Consider the following statements $S1$ and $S2$ about the relational data model:

- $S1$: A relation scheme can have at most one foreign key.
- $S2$: A foreign key in a relation scheme R cannot be used to refer to tuples of $R.$

Which one of the following choices is correct?

A. Both $S1$ and $S2$ are true
B. $S1$ is true and $S2$ is false
C. $S1$ is false and $S2$ is true
D. Both $S1$ and $S2$ are false

gatecse-2021-set2 databases referential-integrity one-mark

Answer key

5.15 Relational Algebra (31)

5.15.1 Relational Algebra: GATE CSE 1992 | Question: 13b

Suppose we have a database consisting of the following three relations:

FREQUENTS	(CUSTOMER, HOTEL)
SERVES	(HOTEL, SNACKS)
LIKES	(CUSTOMER, SNACKS)

The first indicates the hotels each customer visits, the second tells which snacks each hotel serves and last indicates which snacks are liked by each customer. Express the following query in relational algebra:

Print the hotels the serve the snack that customer Rama likes.

gate1992 databases relational-algebra normal descriptive

Answer key

5.15.2 Relational Algebra: GATE CSE 1994 | Question: 13

Consider the following relational schema:

- COURSES (cno, cname)
- STUDENTS (rollno, sname, age, year)
- REGISTERED_FOR (cno, rollno)

The underlined attributes indicate the primary keys for the relations. The ‘year’ attribute for the STUDENTS relation indicates the year in which the student is currently studying (First year, Second year etc.)

A. Write a relational algebra query to print the roll number of students who have registered for cno 322.
B. Write a SQL query to print the age and year of the youngest student in each year.

gate1994 databases relational-algebra sql normal descriptive

Answer key

5.15.3 Relational Algebra: GATE CSE 1994 | Question: 3.8

Give a relational algebra expression using only the minimum number of operators from $(\cup, -)$ which is equivalent to $R \cap S$.

gate1994 databases relational-algebra normal descriptive

Answer key

5.15.4 Relational Algebra: GATE CSE 1995 | Question: 27

Consider the relation scheme.

AUTHOR	(ANAME, INSTITUTION, ACITY, AGE)
PUBLISHER	(PNAME, PCITY)
BOOK	(TITLE, ANAME, PNAME)

Express the following queries using (one or more of) SELECT, PROJECT, JOIN and DIVIDE operations.

A. Get the names of all publishers.
B. Get values of all attributes of all authors who have published a book for the publisher with PNAME='TECHNICAL PUBLISHERS'.
C. Get the names of all authors who have published a book for any publisher located in Madras

gate1995 databases relational-algebra normal descriptive

Answer key

5.15.5 Relational Algebra: GATE CSE 1996 | Question: 27

A library relational database system uses the following schema

- USERS (User#, User Name, Home Town)
- BOOKS (Book#, Book Title, Author Name)
- ISSUED (Book#, User#, Date)

Explain in one English sentence, what each of the following relational algebra queries is designed to determine

a. $\sigma_{\text{User\#}=6} \left(\pi_{\text{User\#, Book Title}} \left((\text{USERS} \bowtie \text{ISSUED}) \bowtie \text{BOOKS}\right)\right)$
b. $\pi_{\text{Author Name}} \left(\text{BOOKS} \bowtie \sigma_{\text{Home Town}=\text{Delhi}} \left(\text{USERS} \bowtie \text{ISSUED}\right)\right)$

gate1996 databases relational-algebra descriptive

Answer key

5.15.6 Relational Algebra: GATE CSE 1997 | Question: 76-a

Consider the following relational database schema:

- EMP (<u>eno</u> name, age)
- PROJ (<u>pno</u> name)
- INVOLVED (<u>eno, pno</u>)

EMP contains information about employees. PROJ about projects and involved about which employees involved in which projects. The underlined attributes are the primary keys for the respective relations.

What is the relational algebra expression containing one or more of $\{\sigma, \pi, \times, \rho, -\}$ which is equivalent to SQL query.

```
select eno  from EMP|INVOLVED where EMP.eno=INVOLVED.eno  and INVOLVED.pno=3
```

gate1997 databases sql relational-algebra descriptive

Answer key☟

5.15.7 Relational Algebra: GATE CSE 1998 | Question: 1.33

Given two union compatible relations $R_1(A, B)$ and $R_2(C, D)$, what is the result of the operation $R_1 \bowtie_{A=C \wedge B=D} R_2$?

A. $R_1 \cup R_2$
B. $R_1 \times R_2$
C. $R_1 - R_2$
D. $R_1 \cap R_2$

gate1998 normal relational-algebra

Answer key☟

5.15.8 Relational Algebra: GATE CSE 1998 | Question: 27

Consider the following relational database schemes:

- COURSES (Cno, Name)
- PRE_REQ(Cno, Pre_Cno)
- COMPLETED (Student_no, Cno)

COURSES gives the number and name of all the available courses.

PRE_REQ gives the information about which courses are pre-requisites for a given course.

COMPLETED indicates what courses have been completed by students

Express the following using relational algebra:

List all the courses for which a student with Student_no 2310 has completed all the pre-requisites.

gate1998 databases relational-algebra normal descriptive

Answer key☟

5.15.9 Relational Algebra: GATE CSE 1999 | Question: 1.18, ISRO2016-53

Consider the join of a relation R with a relation S. If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

A. $m+n$ and 0
B. mn and 0
C. $m+n$ and $|m-n|$
D. mn and $m+n$

gate1999 databases relational-algebra easy isro2016

Answer key☟

5.15.10 Relational Algebra: GATE CSE 2000 | Question: 1.23, ISRO2016-57

Given the relations

- employee (name, salary, dept-no), and
- department (dept-no, dept-name,address),

Which of the following queries cannot be expressed using the basic relational algebra operations $(\sigma, \pi, \times, \bowtie, \cup, \cap, -)$?

A. Department address of every employee
B. Employees whose name is the same as their department name

C. The sum of all employees' salaries
D. All employees of a given department

gatecse-2000 databases relational-algebra easy isro2016

Answer key☟

5.15.11 Relational Algebra: GATE CSE 2001 | Question: 1.24

Suppose the adjacency relation of vertices in a graph is represented in a table Adj (X, Y). Which of the following queries cannot be expressed by a relational algebra expression of constant length?

A. List all vertices adjacent to a given vertex
B. List all vertices which have self loops
C. List all vertices which belong to cycles of less than three vertices
D. List all vertices reachable from a given vertex

gatecse-2001 databases relational-algebra normal

Answer key☟

5.15.12 Relational Algebra: GATE CSE 2001 | Question: 1.25

Let r and s be two relations over the relation schemes R and S respectively, and let A be an attribute in R. The relational algebra expression $\sigma_{A=a}(r \bowtie s)$ is always equal to

A. $\sigma_{A=a}(r)$
B. r
C. $\sigma_{A=a}(r) \bowtie s$
D. None of the above

gatecse-2001 databases relational-algebra

Answer key☟

5.15.13 Relational Algebra: GATE CSE 2002 | Question: 15

A university placement center maintains a relational database of companies that interview students on campus and make job offers to those successful in the interview. The schema of the database is given below:

$\text{COMPANY}(\underline{\text{cname}}, \text{clocation})$	$\text{STUDENT}(\underline{\text{srollno}}, \text{sname}, \text{sdegree})$
$\text{INTERVIEW}(\underline{\text{cname}, \text{srollno}}, \text{idate})$	$\text{OFFER}(\underline{\text{cname}, \text{srollno}}, \text{osalary})$

The COMPANY relation gives the name and location of the company. The STUDENT relation gives the student's roll number, name and the degree program for which the student is registered in the university. The INTERVIEW relation gives the date on which a student is interviewed by a company. The OFFER relation gives the salary offered to a student who is successful in a company's interview. The key for each relation is indicated by the underlined attributes

a. Write a **relational algebra** expressions (using only the operators $\bowtie, \sigma, \pi, \cup, -$) for the following queries.
   i. List the $rollnumbers$ and $names$ of students who attended at least one interview but did not receive any job offer.
   ii. List the $rollnumbers$ and $names$ of students who went for interviews and received job offers from $every$ company with which they interviewed.
b. Write an SQL query to list, for each degree program in which more than $five$ students were offered jobs, the name of the degree and the average offered salary of students in this degree program.

gatecse-2002 databases normal descriptive relational-algebra sql

Answer key☟

5.15.14 Relational Algebra: GATE CSE 2003 | Question: 30

Consider the following SQL query

Select distinct $a_1, a_2, \ldots, a_n$

from $r_1, r_2, \ldots, r_m$
where P
For an arbitrary predicate P, this query is equivalent to which of the following relational algebra expressions?

A. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \times r_2 \times \cdots \times r_m\right)$
B. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \bowtie r_2 \bowtie \cdots \bowtie r_m\right)$
C. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \cup r_2 \cup \cdots \cup r_m\right)$
D. $\Pi_{a_1,a_2,\ldots a_n} \sigma_p \left(r_1 \cap r_2 \cap \cdots \cap r_m\right)$

gatecse-2003 databases relational-algebra normal

Answer key☟

5.15.15 Relational Algebra: GATE CSE 2004 | Question: 51

Consider the relation Student (<u>name</u>, sex, marks), where the primary key is shown underlined, pertaining to students in a class that has at least one boy and one girl. What does the following relational algebra expression produce? (Note: ρ is the rename operator).

$\pi_{name}\{\sigma_{sex=female}(\text{Student})\} - \pi_{name}(\text{Student} \bowtie_{(sex=female \wedge x=male \wedge marks \leq m)} \rho_{n,x,m}(\text{Student}))$

A. names of girl students with the highest marks
B. names of girl students with more marks than some boy student
C. names of girl students with marks not less than some boy student
D. names of girl students with more marks than all the boy students

gatecse-2004 databases relational-algebra normal

Answer key☟

5.15.16 Relational Algebra: GATE CSE 2007 | Question: 59

Information about a collection of students is given by the relation $\text{studInfo}(\underline{\text{studId}}, \text{ name}, \text{sex})$. The relation $\text{enroll}(\text{studId}, \text{ courseId})$ gives which student has enrolled for (or taken) what course(s). Assume that every course is taken by at least one male and at least one female student. What does the following relational algebra expression represent?

$\pi_{courceId}\left((\pi_{\text{studId}}\left(\sigma_{sex='female'}(\text{studInfo})\right) \times \pi_{courseId}(\text{enroll})) - \text{enroll}\right)$

A. Courses in which all the female students are enrolled.
B. Courses in which a proper subset of female students are enrolled.
C. Courses in which only male students are enrolled.
D. None of the above

gatecse-2007 databases relational-algebra normal

Answer key☟

5.15.17 Relational Algebra: GATE CSE 2008 | Question: 68

Let R and S be two relations with the following schema

$R(\underline{P,Q}, R1, R2, R3)$

$S(\underline{P,Q}, S1, S2)$

where $\{P, Q\}$ is the key for both schemas. Which of the following queries are equivalent?

I. $\Pi_P\left(R \bowtie S\right)$
II. $\Pi_P\left(R\right) \bowtie \Pi_P\left(S\right)$
III. $\Pi_P\left(\Pi_{P,Q}\left(R\right) \cap \Pi_{P,Q}\left(S\right)\right)$
IV. $\Pi_P\left(\Pi_{P,Q}\left(R\right) - \left(\Pi_{P,Q}\left(R\right) - \Pi_{P,Q}\left(S\right)\right)\right)$

A. Only I and II B. Only I and III C. Only I, II and III D. Only I, III and IV

gatecse-2008 databases relational-algebra normal

Answer key

5.15.18 Relational Algebra: GATE CSE 2014 Set 3 | Question: 21

What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where $A1, A2$ are sets of attributes in r with $A1 \subset A2$ and $F1, F2$ are Boolean expressions based on the attributes in r?

A. $\pi_{A1}(\sigma_{(F1 \wedge F2)}(r))$
B. $\pi_{A1}(\sigma_{(F1 \vee F2)}(r))$
C. $\pi_{A2}(\sigma_{(F1 \wedge F2)}(r))$
D. $\pi_{A2}(\sigma_{(F1 \vee F2)}(r))$

gatecse-2014-set3 databases relational-algebra easy

Answer key

5.15.19 Relational Algebra: GATE CSE 2014 Set 3 | Question: 30

Consider the relational schema given below, where **eId** of the relation **dependent** is a foreign key referring to **empId** of the relation **employee**. Assume that every employee has at least one associated dependent in the **dependent** relation.

- **employee (empId, empName, empAge)**
- **dependent (depId, eId, depName, depAge)**

Consider the following relational algebra query:

$$\Pi_{empId}(employee) - \Pi_{empId}(employee \bowtie_{(empId = eID) \wedge (empAge \leq depAge)} dependent)$$

The above query evaluates to the set of **empId**s of employees whose age is greater than that of

A. some dependent.
B. all dependents.
C. some of his/her dependents.
D. all of his/her dependents.

gatecse-2014-set3 databases relational-algebra normal

Answer key

5.15.20 Relational Algebra: GATE CSE 2015 Set 1 | Question: 7

SELECT operation in SQL is equivalent to

A. The selection operation in relational algebra
B. The selection operation in relational algebra, except that SELECT in SQL retains duplicates
C. The projection operation in relational algebra
D. The projection operation in relational algebra, except that SELECT in SQL retains duplicates

gatecse-2015-set1 databases sql relational-algebra easy

Answer key

5.15.21 Relational Algebra: GATE CSE 2017 Set 1 | Question: 46

Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below.

StudentName	**CourseName**
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is made on the database.

- $T1 \leftarrow \pi_{CourseName}\ (\sigma_{StudentName=SA}\ (CR))$
- $T2 \leftarrow CR \div T1$

The number of rows in $T2$ is ______________ .

gatecse-2017-set1 databases relational-algebra normal numerical-answers

Answer key

5.15.22 Relational Algebra: GATE CSE 2018 | Question: 41

Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$Q : r \bowtie (\sigma_{B<5}(s))$

Let LOJ denote the natural left outer-join operation. Assume that r and s contain no null values.

Which of the following is NOT equivalent to Q?

A. $\sigma_{B<5}(r \bowtie s)$
B. $\sigma_{B<5}(r\ LOJ\ s)$
C. $r\ LOJ\ (\sigma_{B<5}(s))$
D. $\sigma_{B<5}(r)\ LOJ\ s$

gatecse-2018 databases relational-algebra normal two-marks

Answer key

5.15.23 Relational Algebra: GATE CSE 2019 | Question: 55

Consider the following relations $P(X, Y, Z), Q(X, Y, T)$ and $R(Y, V)$.

Table: P

X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Table: Q

X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

Table: R

Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?

$$\Pi_x(\sigma_{(P.Y=R.Y \wedge R.V=V2))}\ (P \times R)) - \Pi_x(\sigma_{(Q.Y=R.Y \wedge Q.T>2))}\ (Q \times R))$$

Answer: ________

gatecse-2019 numerical-answers databases relational-algebra two-marks

Answer key

5.15.24 Relational Algebra: GATE CSE 2021 Set 1 | Question: 27

The following relation records the age of 500 employees of a company, where $empNo$ (indicating the employee number) is the key:

$$empAge(\underline{empNo}, age)$$

Consider the following relational algebra expression:

$$\Pi_{empNo}(empAge \bowtie_{(age>age1)}\ \rho_{empNo1,age1}(empAge))$$

What does the above expression generate?

A. Employee numbers of only those employees whose age is the maximum
B. Employee numbers of only those employees whose age is more than the age of exactly one other employee
C. Employee numbers of all employees whose age is not the minimum
D. Employee numbers of all employees whose age is the minimum

gatecse-2021-set1 databases relational-algebra two-marks

Answer key

5.15.25 Relational Algebra: GATE CSE 2022 | Question: 15

Consider the following three relations in a relational database.

$$\text{Employee}(\underline{\text{eId}}, \text{Name}),\ \text{Brand}(\underline{\text{bId}}, \text{bName}),\ \text{Own}(\underline{\text{eId}}, \underline{\text{bId}})$$

Which of the following relational algebra expressions return the set of $eIds$ who own all the brands?

A. $\Pi_{eId}\,(\Pi_{eId,bId}(Own)/\Pi_{bId}(Brand))$
B. $\Pi_{eId}(Own) - \Pi_{eId}\,((\Pi_{eId}(Own) \times \Pi_{bId}(Brand)) - \Pi_{eId,bId}(Own))$
C. $\Pi_{eId}\,(\Pi_{eId,bId}(Own)/\Pi_{bId}(Own))$
D. $\Pi_{eId}\,((\Pi_{eId}(Own) \times \Pi_{bId}(Own))/\Pi_{bId}(Brand))$

gatecse-2022 databases relational-algebra multiple-selects one-mark

Answer key

5.15.26 Relational Algebra: GATE CSE 2024 | Set 1 | Question: 25

Consider the following two relations, $R(A, B)$ and $S(A, C)$:

R	
A	**B**
10	20
20	30
30	40
30	50
50	95

S	
A	**C**
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$\sigma_{B<C}\left(R \bowtie_{R.A=S.A} S\right)$ is ___________.

gatecse2024-set1 numerical-answers databases relational-algebra one-mark

Answer key

5.15.27 Relational Algebra: GATE CSE 2024 | Set 2 | Question: 35

The relation schema, Person (pid, city), describes the city of residence for every person uniquely identified by pid. The following relational algebra operators are available: selection, projection, cross product, and rename.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is

A. 1 B. 2 C. 3 D. 4

gatecse2024-set2 databases relational-algebra two-marks

Answer key

5.15.28 Relational Algebra: GATE DA 2025 | Question: 52

Consider the following tables, Loan and Borrower, of a bank.

Loan		
loan_number	**branch_name**	**amount**
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

Borrower	
customer_name	**loan_num**
Anand	L11
Karteek	L11
Karteek	L14
Ankita	L15
Gopal	L19
Karteek	L22
Karteek	L23
Sunil	L23
Sunil	L25

Query: $\pi_{\text{branch\_name, customer\_name}}$ (**Loan** $\bowtie$ **Borrower**) $\div \pi_{\text{branch\_name}}$ (**Loan**) where $\bowtie$ denotes natural join.

The number of tuples returned by the above relational algebra query is ___________ (Answer in integer)

gateda-2025 databases relational-algebra numerical-answers two-marks

Answer key

5.15.29 Relational Algebra: GATE DA 2025 | Question: 7

Consider the following three relations:

$\text{Car (model, year, }\underline{\text{serial}}\text{, color)}$
$\text{Make (maker, }\underline{\text{model}}\text{)}$
$\text{Own (}\underline{\text{owner}}\text{, }\underline{\text{serial}}\text{)}$

A tuple in Car represents a specific car of a given model, made in a given year, with a serial number and a color. A tuple in Make specifies that a maker company makes cars of a certain model. A tuple in Own specifies that an owner owns the car with a given serial number. Keys are underlined; ($\underline{\text{owner}}$, $\underline{\text{serial}}$) together form key for Own. ($\bowtie$ denotes natural join)

$$\pi_{\text{owner}}\ (\text{Own} \bowtie (\sigma_{\text{color=''red''}}\ (\text{Car} \bowtie (\sigma_{\text{maker=''ABC''}}\ \text{Make}))))$$

Which one of the following options describes what the above expression computes?

A. All owners of a red car, a car made by ABC, or a red car made by ABC
B. All owners of more than one car, where at least one car is red and made by ABC
C. All owners of a red car made by ABC
D. All red cars made by ABC

gateda-2025 databases relational-algebra one-mark

Answer key☟

5.15.30 Relational Algebra: GATE DS&AI 2024 | Question: 16

Consider a database that includes the following relations:

Defender(name, rating, side, goals)
Forward(name, rating, assists, goals)
Team(name, club, price)

Which ONE of the following relational algebra expressions checks that every name occurring in Team appears in either Defender or Forward, where ϕ denotes the empty set?

A. Π_{name} (Team)$\setminus$ (Π_{name} (Defender)$\cap\Pi_{\text{name}}$ (Forward)) $= \phi$
B. (Π_{name} (Defender)$\cap\Pi_{\text{name}}$ (Forward)) $\setminus\Pi_{\text{name}}$ (Team)$= \phi$
C. Π_{name} (Team)$\setminus$ (Π_{name} (Defender)$\cup\Pi_{\text{name}}$ (Forward)) $= \phi$
D. (Π_{name} (Defender)$\cup\Pi_{\text{name}}$ (Forward)) $\setminus\Pi_{\text{name}}$ (Team)$= \phi$

gate-ds-ai-2024 relational-algebra databases one-mark

Answer key☟

5.15.31 Relational Algebra: GATE IT 2005 | Question: 68

A table 'student' with schema (roll, name, hostel, marks), and another table 'hobby' with schema (roll, hobbyname) contains records as shown below:

Table: student

Roll	Name	Hostel	Marks
1798	Manoj Rathor	7	95
2154	Soumic Banerjee	5	68
2369	Gumma Reddy	7	86
2581	Pradeep pendse	6	92
2643	Suhas Kulkarni	5	78
2711	Nitin Kadam	8	72
2872	Kiran Vora	5	92
2926	Manoj Kunkalikar	5	94
2959	Hemant Karkhanis	7	88
3125	Rajesh Doshi	5	82

Table: hobby

Roll	Hobby Name
1798	chess
1798	music
2154	music
2369	swimming
2581	cricket
2643	chess
2643	hockey
2711	volleyball
2872	football
2926	cricket
2959	photography
3125	music
3125	chess

The following SQL query is executed on the above tables:

```
select hostel
from student natural join hobby
where marks >= 75 and roll between 2000 and 3000;
```

Relations S and H with the same schema as those of these two tables respectively contain the same information as tuples. A new relation S' is obtained by the following relational algebra operation:

$$S' = \Pi_{\text{hostel}}((\sigma_{s.roll=H.roll}(\sigma_{marks>75 \text{ and } roll>2000 \text{ and } roll<3000}(S)) \times (H))$$

The difference between the number of rows output by the SQL statement and the number of tuples in S' is

A. 6 B. 4 C. 2 D. 0

gateit-2005 databases sql relational-algebra normal

Answer key

5.16 Relational Calculus (15)

5.16.1 Relational Calculus: GATE CSE 1993 | Question: 23

The following relations are used to store data about students, courses, enrollment of students in courses and teachers of courses. Attributes for primary key in each relation are marked by '*'.

```
Students (rollno*, sname, saddr)
courses (cno*, cname)
enroll(rollno*, cno*, grade)
teach(tno*, tname, cao*)
```

(cno is course number cname is course name, tno is teacher number, tname is teacher name, sname is student name, etc.)

Write a SQL query for retrieving roll number and name of students who got A grade in at least one course taught by teacher names Ramesh for the above relational database.

gate1993 databases sql relational-calculus normal descriptive

Answer key

5.16.2 Relational Calculus: GATE CSE 1993 | Question: 24

The following relations are used to store data about students, courses, enrollment of students in courses and teachers of courses. Attributes for primary key in each relation are marked by '*'.

- $\text{students(rollno*, sname, saddr)}$
- $\text{courses(cno*, cname)}$
- $\text{enroll(rollno*, cno*, grade)}$
- $\text{teach(tno*, tname, cao*)}$

(cno is course number, cname is course name, tno is teacher number, tname is teacher name, sname is student name, etc.)

For the relational database given above, the following functional dependencies hold:

- $\text{rollno} \rightarrow \text{sname, saddr}$
- $\text{cno} \rightarrow \text{cname}$
- $\text{tno} \rightarrow \text{tname}$
- $\text{rollno, cno} \rightarrow \text{grade}$

a. Is the database in 3^{rd} normal form $(3NF)$?
b. If yes, prove that it is in 3NF. If not, normalize the relations so that they are in 3NF (without proving).

gate1993 databases sql relational-calculus normal descriptive

Answer key

5.16.3 Relational Calculus: GATE CSE 1998 | Question: 2.19

Which of the following query transformations (i.e., replacing the l.h.s. expression by the r.h.s expression) is incorrect? R1 and R2 are relations, C1 and C2 are selection conditions and A1 and A2 are attributes of R1.

A. $\sigma_{C_1}\left(\sigma_{C_2}\left(R_1\right)\right) \rightarrow \sigma_{C_2}\left(\sigma_{C_1}\left(R_1\right)\right)$
B. $\sigma_{C_1}\left(\pi_{A_1}\left(R_1\right)\right) \rightarrow \pi_{A_1}\left(\sigma_{C_1}\left(R_1\right)\right)$
C. $\sigma_{C_1}\left(R_1 \cup R_2\right) \rightarrow \sigma_{C_1}\left(R_1\right) \cup \sigma_{C_1}\left(R_2\right)$
D. $\pi_{A_1}\left(\sigma_{C_1}\left(R_1\right)\right) \rightarrow \sigma_{C_1}\left(\pi_{A_1}\left(R_1\right)\right)$

gate1998 databases relational-calculus normal

Answer key

5.16.4 Relational Calculus: GATE CSE 1999 | Question: 1.19

The relational algebra expression equivalent to the following tuple calculus expression:

$\{t \mid t \in r \wedge (t[A] = 10 \wedge t[B] = 20)\}$ is

A. $\sigma_{(A=10 \vee B=20)}(r)$
B. $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
C. $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
D. $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

gate1999 databases relational-calculus normal

Answer key

5.16.5 Relational Calculus: GATE CSE 2001 | Question: 2.24

Which of the following relational calculus expression is not safe?

A. $\{t \mid \exists u \in R_1\left(t[A] = u[A]\right) \wedge \neg \exists s \in R_2\left(t[A] = s[A]\right)\}$
B. $\{t \mid \forall u \in R_1\left(u[A] =" x " \Rightarrow \exists s \in R_2\left(t[A] = s[A] \wedge s[A] = u[A]\right)\right)\}$
C. $\{t \mid \neg(t \in R_1)\}$
D. $\{t \mid \exists u \in R_1\left(t[A] = u[A]\right) \wedge \exists s \in R_2\left(t[A] = s[A]\right)\}$

gatecse-2001 relational-calculus normal databases

Answer key

5.16.6 Relational Calculus: GATE CSE 2002 | Question: 1.20

With regards to the expressive power of the formal relational query languages, which of the following statements is true?

A. Relational algebra is more powerful than relational calculus
B. Relational algebra has the same power as relational calculus
C. Relational algebra has the same power as safe relational calculus
D. None of the above

gatecse-2002 databases relational-calculus normal

Answer key

5.16.7 Relational Calculus: GATE CSE 2004 | Question: 13

Let $R_1\left(\underline{A}, B, C\right)$ and $R_2\left(\underline{D}, E\right)$ be two relation schema, where the primary keys are shown underlined, and let C be a foreign key in R_1 referring to R_2. Suppose there is no violation of the above referential integrity constraint in the corresponding relation instances r_1 and r_2. Which of the following relational algebra expressions would necessarily produce an empty relation?

A. $\Pi_D(r_2) - \Pi_C(r_1)$
B. $\Pi_C(r_1) - \Pi_D(r_2)$
C. $\Pi_D\left(r_1 \bowtie_{C \neq D} r_2\right)$
D. $\Pi_C\left(r_1 \bowtie_{C = D} r_2\right)$

gatecse-2004 databases relational-calculus easy

Answer key

5.16.8 Relational Calculus: GATE CSE 2007 | Question: 60

Consider the relation **employee**(name, sex, supervisorName) with *name* as the key, *supervisorName* gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$$\{e.name \mid employee(e) \wedge (\forall x) \left[\neg employee\left(x\right) \vee x.supervisorName \neq e.name \vee x.sex = \text{``}male\text{''}\right]\}$$

A. Names of employees with a male supervisor.
B. Names of employees with no immediate male subordinates.
C. Names of employees with no immediate female subordinates.
D. Names of employees with a female supervisor.

gatecse-2007 databases relational-calculus normal

Answer key

5.16.9 Relational Calculus: GATE CSE 2008 | Question: 15

Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r\left(P\left(t\right)\right)$?

I. $\neg \exists t \in r\left(P\left(t\right)\right)$
II. $\exists t \notin r\left(P\left(t\right)\right)$
III. $\neg \exists t \in r\left(\neg P\left(t\right)\right)$
IV. $\exists t \notin r\left(\neg P\left(t\right)\right)$

A. I only
B. II only
C. III only
D. III and IV only

gatecse-2008 databases relational-calculus normal

Answer key

5.16.10 Relational Calculus: GATE CSE 2009 | Question: 45

Let R and S be relational schemes such that $R = \{a, b, c\}$ and $S = \{c\}$. Now consider the following queries on the database:

1. $\pi_{R-S}(r) - \pi_{R-S}\left(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r)\right)$
2. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s\left(\exists v \in r\left(u = v[S] \wedge t = v\left[R - S\right]\right)\right)\}$

3. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r\,(\exists u \in s\,(u = v[S] \wedge t = v\,[R-S]))\}$

4.
```
Select R.a,R.b
  From R,S
  Where R.c = S.c
```

Which of the above queries are equivalent?

A. 1 and 2 B. 1 and 3 C. 2 and 4 D. 3 and 4

gatecse-2009 databases relational-calculus difficult

Answer key

5.16.11 Relational Calculus: GATE CSE 2013 | Question: 35

Consider the following relational schema.

- Students(rollno: integer, sname: string)
- Courses(courseno: integer, cname: string)
- Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

"Find the distinct names of all students who score more than 90% in the course numbered 107"

I.
```
SELECT DISTINCT S.sname FROM Students as S, Registration
  as R WHERE
  R.rollno=S.rollno AND R.courseno=107 AND R.percent >90
```

II. $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}\,(Registration \bowtie Students))$

III. $\{T \mid \exists S \in Students, \exists R \in Registration(S.rollno = R.rollno$
$\wedge R.courseno = 107 \wedge R.percent > 90 \wedge T.sname = S.sname)\}$

IV.
$$\{\langle S_N \rangle \mid \exists S_R \exists R_P (\langle S_R, S_N \rangle \in Students \wedge$$
$$\langle S_R, 107, R_P \rangle \in Registration \wedge R_P > 90)\}$$

A. I, II, III and IV B. I, II and III only
C. I, II and IV only D. II, III and IV only

gatecse-2013 databases sql relational-calculus normal

Answer key

5.16.12 Relational Calculus: GATE CSE 2017 Set 1 | Question: 41

Consider a database that has the relation schemas EMP(EmpId, EmpName, DeptId), and DEPT(DeptName, DeptId). Note that the DeptId can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

I. $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$
II. $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$
III. $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] = v[\text{DeptId}]))\}$

Which of the above queries are safe?

A. I and II only B. I and III only C. II and III only D. I, II and III

gatecse-2017-set1 databases relational-calculus safe-query normal

Answer key

5.16.13 Relational Calculus: GATE IT 2006 | Question: 15

Which of the following relational query languages have the same expressive power?

I. Relational algebra
II. Tuple relational calculus restricted to safe expressions
III. Domain relational calculus restricted to safe expressions

A. II and III only
B. I and II only
C. I and III only
D. I, II and III

gateit-2006 databases relational-algebra relational-calculus easy

Answer key

5.16.14 Relational Calculus: GATE IT 2007 | Question: 65

Consider a selection of the form $\sigma_{A \leq 100}(r)$, where r is a relation with 1000 tuples. Assume that the attribute values for A among the tuples are uniformly distributed in the interval $[0, 500]$. Which one of the following options is the best estimate of the number of tuples returned by the given selection query ?

A. 50
B. 100
C. 150
D. 200

gateit-2007 databases relational-calculus probability normal

Answer key

5.16.15 Relational Calculus: GATE IT 2008 | Question: 75

Consider the following relational schema:

- $\text{Student}(\underline{\text{school-id}, \text{sch-roll-no}}, \text{sname}, \text{saddress})$
- $\text{School}(\underline{\text{school-id}}, \text{sch-name}, \text{sch-address}, \text{sch-phone})$
- $\text{Enrolment}(\underline{\text{school-id}, \text{sch-roll-no}}, \text{erollno}, \text{examname})$
- $\text{ExamResult}(\underline{\text{erollno}, \text{examname}}, \text{marks})$

Consider the following tuple relational calculus query.

$$\{t \mid \exists E \in \text{Enrolment} \;\; t = E.\text{school-id} \wedge |\{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult} \;\; B.\text{er}$$

If a student needs to score more than 35 marks to pass an exam, what does the query return?

A. The empty set
B. schools with more than 35% of its students enrolled in some exam or the other
C. schools with a pass percentage above 35% over all exams taken together
D. schools with a pass percentage above 35% over each exam

gateit-2008 databases relational-calculus normal

Answer key

5.17 Relational Model (2)

5.17.1 Relational Model: GATE CSE 2023 | Question: 6

Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?

A. Number of attributes of its relation schema.
B. Number of tuples stored in the relation.
C. Number of entries in the relation.
D. Number of distinct domains of its relation schema.

gatecse-2023 databases relational-model one-mark easy

Answer key

5.17.2 Relational Model: GATE DA 2025 | Question: 46

Consider the following two relations, named Customer and Person, in a database:

```
Person (
aadhaar CHAR(12) PRIMARY KEY,
name VARCHAR(32));

Customer (
name VARCHAR (32),
```

```
  email VARCHAR(32) PRIMARY KEY,
  phone CHAR(10),
  aadhaar CHAR(12),
    FOREIGN KEY (aadhaar) REFERENCES Person(aadhaar));
```

Which of the following statements is/are correct?

A. aadhaar is a candidate key in the Customer relation
B. phone can be NULL in the Customer relation
C. aadhaar is a candidate key in the Person relation
D. aadhaar can be NULL in the Person relation

gateda-2025 databases relational-model multiple-selects two-marks

Answer key

5.18 SQL (58)

5.18.1 SQL: GATE CSE 1988 | Question: 12iii

Describe the relational algebraic expression giving the relation returned by the following SQL query.

```
Select    SNAME
from      S
Where     SNOin
          (select    SNO
          from       SP
          where      PNOin
                     (select    PNO
                     from       P
                     Where      COLOUR='BLUE'))
```

gate1988 normal descriptive databases sql

Answer key

5.18.2 SQL: GATE CSE 1988 | Question: 12iv

```
Select    SNAME
from      S
Where     SNOin
          (select    SNO
          from       SP
          where      PNOin
                     (select    PNO
                     from       P
                     Where      COLOUR='BLUE'))
```

What relations are being used in the above SQL query? Given at least two attributes of each of these relations.

gate1988 normal descriptive databases sql

Answer key

5.18.3 SQL: GATE CSE 1990 | Question: 10-a

Consider the following relational database:

- employees (eno, ename, address, basic-salary)
- projects (pno, pname, nos-of-staffs-allotted)
- working (pno, eno, pjob)

The queries regarding data in the above database are formulated below in SQL. Describe in ENGLISH sentences the two queries that have been posted:

i.
```
SELECT ename
FROM employees
WHERE eno IN
  (SELECT eno
  FROM working
  GROUP BY eno
```

```
HAVING COUNT(*)=
   (SELECT COUNT(*)
   FROM projects))
```

ii.
```
SELECT pname
FROM projects
WHERE pno IN
   (SELECT pno
   FROM projects
   MINUS
   SELECT DISTINCT pno
   FROM working);
```

gate1990 descriptive databases sql

Answer key

5.18.4 SQL: GATE CSE 1991 | Question: 12,b

Suppose a database consist of the following relations:

```
SUPPLIER (SCODE,SNAME,CITY).

PART (PCODE,PNAME,PDESC,CITY).

PROJECTS (PRCODE,PRNAME,PRCITY).

SPPR (SCODE,PCODE,PRCODE,QTY).
```

Write algebraic solution to the following :

i. Get SCODE values for suppliers who supply to both projects PR1 and PR2.
ii. Get PRCODE values for projects supplied by at least one supplier not in the same city.

sql gate1991 normal databases descriptive

Answer key

5.18.5 SQL: GATE CSE 1991 | Question: 12-a

Suppose a database consist of the following relations:

```
SUPPLIER (SCODE,SNAME,CITY).
PART (PCODE,PNAME,PDESC,CITY).
PROJECTS (PRCODE,PRNAME,PRCITY).
SPPR (SCODE,PCODE,PRCODE,QTY).
```

Write SQL programs corresponding to the following queries:

i. Print PCODE values for parts supplied to any project in DEHLI by a supplier in DELHI.
ii. Print all triples <CITY, PCODE, CITY> such that a supplier in first city supplies the specified part to a project in the second city, but do not print the triples in which the two CITY values are same.

gate1991 databases sql normal descriptive

Answer key

5.18.6 SQL: GATE CSE 1998 | Question: 7-a

Suppose we have a database consisting of the following three relations.

- $\text{FREQUENTS (student, parlor)}$ giving the parlors each student visits.
- $\text{SERVES (parlor, ice-cream)}$ indicating what kind of ice-creams each parlor serves.
- $\text{LIKES (student, ice-cream)}$ indicating what ice-creams each student likes.

(Assume that each student likes at least one ice-cream and frequents at least one parlor)

Express the following in SQL:

Print the students that frequent at least one parlor that serves some ice-cream that they like.

gate1998 databases sql descriptive

Answer key

5.18.7 SQL: GATE CSE 1999 | Question: 2.25

Which of the following is/are correct?

A. An SQL query automatically eliminates duplicates
B. An SQL query will not work if there are no indexes on the relations
C. SQL permits attribute names to be repeated in the same relation
D. None of the above

gate1999 databases sql easy

Answer key

5.18.8 SQL: GATE CSE 1999 | Question: 22-a

Consider the set of relations

- EMP (Employee-no. Dept-no, Employee-name, Salary)
- DEPT (Dept-no. Dept-name, Location)

Write an SQL query to:

a. Find all employees names who work in departments located at ‘Calcutta’ and whose salary is greater than Rs.50,000.
b. Calculate, for each department number, the number of employees with a salary greater than Rs. 1,00,000.

gate1999 databases sql easy descriptive

Answer key

5.18.9 SQL: GATE CSE 1999 | Question: 22-b

Consider the set of relations

- EMP (Employee-no. Dept-no, Employee-name, Salary)
- DEPT (Dept-no. Dept-name, Location)

Write an SQL query to:

Calculate, for each department number, the number of employees with a salary greater than Rs. 1,00,000

gate1999 databases sql descriptive easy

Answer key

5.18.10 SQL: GATE CSE 2000 | Question: 2.25

Given relations r(w, x) and s(y, z) the result of

select distinct w, x
from r, s

is guaranteed to be same as r, provided.

A. r has no duplicates and s is non-empty
B. r and s have no duplicates
C. s has no duplicates and r is non-empty
D. r and s have the same number of tuples

gatecse-2000 databases sql

Answer key

5.18.11 SQL: GATE CSE 2000 | Question: 2.26

In SQL, relations can contain null values, and comparisons with null values are treated as unknown. Suppose all comparisons with a null value are treated as false. Which of the following pairs is not equivalent?

A. $x = 5 \quad not(not(x = 5))$
B. $x = 5 \quad x > 4$ and $x < 6$, where x is an integer
C. $x \neq 5 \quad not(x = 5)$
D. none of the above

gatecse-2000 databases sql normal

Answer key

5.18.12 SQL: GATE CSE 2000 | Question: 22

Consider a bank database with only one relation

transaction (transno, acctno, date, amount)

The amount attribute value is positive for deposits and negative for withdrawals.

a. Define an SQL view TP containing the information
(acctno,T1.date,T2.amount)
for every pair of transaction T1,T2 and such that T1 and T2 are transaction on the same account and the date of T2 is $\leq$ the date of T1.
b. Using only the above view TP, write a query to find for each account the minimum balance it ever reached (not including the 0 balance when the account is created). Assume there is at most one transaction per day on each account and each account has at least one transaction since it was created. To simplify your query, break it up into 2 steps by defining an intermediate view V.

gatecse-2000 databases sql normal descriptive

Answer key

5.18.13 SQL: GATE CSE 2001 | Question: 2.25

Consider a relation geq which represents "greater than or equal to", that is, $(x, y) \in$ geq only if $y \geq x$.

```
create table geq
(
   ib integer not null,
   ub integer not null,
   primary key ib,
   foreign key (ub) references geq on delete cascade
);
```

Which of the following is possible if tuple (x,y) is deleted?

A. A tuple (z,w) with z > y is deleted
B. A tuple (z,w) with z > x is deleted
C. A tuple (z,w) with w < x is deleted
D. The deletion of (x,y) is prohibited

gatecse-2001 databases sql normal

Answer key

5.18.14 SQL: GATE CSE 2001 | Question: 21-a

Consider a relation examinee (regno, name, score), where regno is the primary key to score is a real number.

Write a relational algebra using $(\Pi, \sigma, \rho, \times)$ to find the list of names which appear more than once in examinee.

gatecse-2001 databases sql normal descriptive

Answer key

5.18.15 SQL: GATE CSE 2001 | Question: 21-b

Consider a relation examinee (regno, name, score), where regno is the primary key to score is a real number.

Write an SQL query to list the *regno* of examinees who have a score greater than the average score.

gatecse-2001 databases sql normal descriptive

Answer key

5.18.16 SQL: GATE CSE 2001 | Question: 21-c

Consider a relation $\text{examinee (regno, name, score)}$, where regno is the primary key to score is a real number.

Suppose the relation $\text{appears (regno, centr\_code)}$ specifies the center where an examinee appears. Write an SQL query to list the centr_code having an examinee of score greater than 80.

gatecse-2001 databases sql normal descriptive

Answer key

5.18.17 SQL: GATE CSE 2003 | Question: 86

Consider the set of relations shown below and the SQL query that follows.

Students: (Roll_number, Name, Date_of_birth)

Courses: (Course_number, Course_name, Instructor)

Grades: (Roll_number, Course_number, Grade)

```
Select distinct Name
from Students, Courses, Grades
where Students.Roll_number=Grades.Roll_number
 and Courses.Instructor = 'Korth'
 and Courses.Course_number = Grades.Course_number
 and Grades.Grade = 'A'
```

Which of the following sets is computed by the above query?

A. Names of students who have got an A grade in all courses taught by Korth
B. Names of students who have got an A grade in all courses
C. Names of students who have got an A grade in at least one of the courses taught by Korth
D. None of the above

gatecse-2003 databases sql easy

Answer key

5.18.18 SQL: GATE CSE 2004 | Question: 53

The employee information in a company is stored in the relation

- Employee (name, sex, salary, deptName)

Consider the following SQL query

```
Select deptName
   From Employee
   Where sex = 'M'
   Group by deptName
   Having avg(salary) >
      (select avg (salary) from Employee)
```

It returns the names of the department in which

A. the average salary is more than the average salary in the company
B. the average salary of male employees is more than the average salary of all male employees in the company

C. the average salary of male employees is more than the average salary of employees in same the department
D. the average salary of male employees is more than the average salary in the company

gatecse-2004 databases sql normal

Answer key

5.18.19 SQL: GATE CSE 2005 | Question: 77, ISRO2016-55

The relation **book** (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
select title
from book as B
where (select count(*)
  from book as T
  where T.price>B.price) < 5
```

A. Titles of the four most expensive books
B. Title of the fifth most inexpensive book
C. Title of the fifth most expensive book
D. Titles of the five most expensive books

gatecse-2005 databases sql easy isro2016

Answer key

5.18.20 SQL: GATE CSE 2006 | Question: 67

Consider the relation account (customer, balance) where the customer is a primary key and there are no null values. We would like to rank customers according to decreasing balance. The customer with the largest balance gets rank 1. Ties are not broke but ranks are skipped: if exactly two customers have the largest balance they each get rank 1 and rank 2 is not assigned.

Query1:

```
select A.customer, count(B.customer)
from account A, account B
where A.balance <=B.balance
group by A.customer
```

Query2:

```
select A.customer, 1+count(B.customer)
from account A, account B
where A.balance < B.balance
group by A.customer
```

Consider these statements about Query1 and Query2.

1. Query1 will produce the same row set as Query2 for some but not all databases.
2. Both Query1 and Query 2 are a correct implementation of the specification
3. Query1 is a correct implementation of the specification but Query2 is not
4. Neither Query1 nor Query2 is a correct implementation of the specification
5. Assigning rank with a pure relational query takes less time than scanning in decreasing balance order assigning ranks using ODBC.

Which two of the above statements are correct?

A. 2 and 5 B. 1 and 3 C. 1 and 4 D. 3 and 5

gatecse-2006 databases sql normal

Answer key

5.18.21 SQL: GATE CSE 2006 | Question: 68

Consider the relation enrolled (student, course) in which (student, course) is the primary key, and the relation paid (student, amount) where student is the primary key. Assume no null values and no foreign keys or

integrity constraints.
Given the following four queries:

Query1:

```
select student from enrolled where student in (select student from paid)
```

Query2:

```
select student from paid where student in (select student from enrolled)
```

Query3:

```
select E.student from enrolled E, paid P where E.student = P.student
```

Query4:

```
select student from paid where exists
   (select * from enrolled where enrolled.student = paid.student)
```

Which one of the following statements is correct?

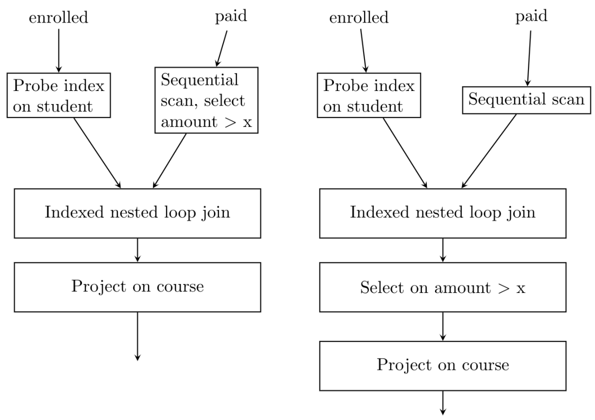
A. All queries return identical row sets for any database
B. Query2 and Query4 return identical row sets for all databases but there exist databases for which Query1 and Query2 return different row sets
C. There exist databases for which Query3 returns strictly fewer rows than Query2
D. There exist databases for which Query4 will encounter an integrity violation at runtime

gatecse-2006 databases sql normal

Answer key

5.18.22 SQL: GATE CSE 2006 | Question: 69

Consider the relation enrolled (student, course) in which (student, course) is the primary key, and the relation paid (student, amount) where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Assume that amounts $6000, 7000, 8000, 9000$ and 10000 were each paid by 20% of the students. Consider these query plans (Plan 1 on left, Plan 2 on right) to *"list all courses taken by students who have paid more than x"*



A disk seek takes $4ms$, disk data transfer bandwidth is 300 MB/s and checking a tuple to see if amount is greater than x takes $10\mu s$. Which of the following statements is correct?

A. Plan 1 and Plan 2 will not output identical row sets for all databases
B. A course may be listed more than once in the output of Plan 1 for some databases
C. For $x = 5000$, Plan 1 executes faster than Plan 2 for all databases
D. For $x = 9000$, Plan I executes slower than Plan 2 for all databases

gatecse-2006 databases sql normal

Answer key

5.18.23 SQL: GATE CSE 2007 | Question: 61

Consider the table **employee**(empId, name, department, salary) and the two queries Q_1, Q_2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is **TRUE** for any arbitrary employee table?

Q_1 :
```
Select e.empId
From employee e
Where not exists
   (Select * From employee s Where s.department = "5" and s.salary >= e.salary)
```

Q_2 :
```
Select e.empId
From employee e
Where e.salary > Any
   (Select distinct salary From employee s Where s.department = "5")
```

A. Q_1 is the correct query
B. Q_2 is the correct query
C. Both Q_1 and Q_2 produce the same answer
D. Neither Q_1 nor Q_2 is the correct query

gatecse-2007 databases sql normal verbal-aptitude

Answer key

5.18.24 SQL: GATE CSE 2009 | Question: 55

Consider the following relational schema:

$\text{Suppliers}(\underline{\text{sid:integer}}, \text{ sname:string, city:string, street:string})$

$\text{Parts}(\underline{\text{pid:integer}}, \text{ pname:string, color:string})$

$\text{Catalog}(\underline{\text{sid:integer, pid:integer}}, \text{cost:real})$

Consider the following relational query on the above database:

```
SELECT S.sname
FROM    Suppliers S
WHERE S.sid NOT IN (SELECT C.sid
                FROM Catalog C
                WHERE C.pid NOT IN (SELECT P.pid
                                FROM Parts P
                                WHERE P.color<>'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. Find the names of all suppliers who have supplied a non-blue part.
B. Find the names of all suppliers who have not supplied a non-blue part.
C. Find the names of all suppliers who have supplied only non-blue part.
D. Find the names of all suppliers who have not supplied only blue parts.

gatecse-2009 databases sql normal

Answer key

5.18.25 SQL: GATE CSE 2009 | Question: 56

Consider the following relational schema:

- $\text{Suppliers}(\underline{\text{sid:integer}}, \text{ sname:string, city:string, street:string})$
- $\text{Parts}(\underline{\text{pid:integer}}, \text{ pname:string, color:string})$
- $\text{Catalog}(\underline{\text{sid:integer, pid:integer}}, \text{cost:real})$

Assume that, in the suppliers relation above, each supplier and each street within a city has unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema?

A. The schema is in BCNF

B. The schema is in 3NF but not in BCNF

C. The schema is in 2NF but not in 3NF

D. The schema is not in 2NF

gatecse-2009 databases sql database-normalization normal

Answer key

5.18.26 SQL: GATE CSE 2010 | Question: 19

A relational schema for a train reservation database is given below.

- **passenger(pid, pname, age)**
- **reservation(pid, class, tid)**

Passenger

pid	pname	Age
0	Sachine	65
1	Rahul	66
2	Sourav	67
3	Anil	69

Reservation

pid	class	tid
0	AC	8200
1	AC	8201
2	SC	8201
5	AC	8203
1	SC	8204
3	AC	8202

What **pids** are returned by the following SQL query for the above instance of the tables?

```
SELECT pid
FROM Reservation
WHERE class='AC' AND
   EXISTS (SELECT *
         FROM Passenger
         WHERE age>65 AND
         Passenger.pid=Reservation.pid)
```

A. $1,0$

B. $1,2$

C. $1,3$

D. $1,5$

gatecse-2010 databases sql normal

Answer key

5.18.27 SQL: GATE CSE 2011 | Question: 32

Consider a database table T containing two columns X and Y each of type integer. After the creation of the table, one record $(\text{X}=1, \text{Y}=1)$ is inserted in the table.

Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being $\text{MX}+1, 2*\text{MY}+1$ respectively. It may be noted that each time after the insertion, values of MX and MY change.

What will be the output of the following SQL query after the steps mentioned above are carried out?

```
SELECT Y FROM T WHERE X=7;
```

A. 127

B. 255

C. 129

D. 257

gatecse-2011 databases sql normal

Answer key

5.18.28 SQL: GATE CSE 2011 | Question: 46

Database table by name Loan_Records is given below.

Borrower	Bank_Manager	Loan_Amount
Ramesh	Sunderajan	10000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	7000.00

What is the output of the following SQL query?

```
SELECT count(*)
FROM (
  SELECT Borrower, Bank_Manager FROM Loan_Records) AS S
  NATURAL JOIN
  (SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T
);
```

A. 3 B. 9 C. 5 D. 6

gatecse-2011 databases sql normal

Answer key

5.18.29 SQL: GATE CSE 2012 | Question: 15

Which of the following statements are **TRUE** about an SQL query?
P : An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause
Q : An SQL query can contain a HAVING clause only if it has a GROUP BY clause
R : All attributes used in the GROUP BY clause must appear in the SELECT clause
S : Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

A. P and R B. P and S C. Q and R D. Q and S

gatecse-2012 databases easy sql ambiguous

Answer key

5.18.30 SQL: GATE CSE 2012 | Question: 51

Consider the following relations A, B and C :

A

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

Id	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > ALL (SELECT B.Age
          FROM B
          WHERE B.Name = 'Arun')
```

A. 4 B. 3 C. 0 D. 1

gatecse-2012 databases sql normal

Answer key

5.18.31 SQL: GATE CSE 2014 Set 1 | Question: 22

Given the following statements:

S1: A foreign key declaration can always be replaced by an equivalent check assertion in SQL.

S2: Given the table $R(a, b, c)$ where a and b together form the primary key, the following is a valid table definition.

```
CREATE TABLE S (
    a INTEGER,
    d INTEGER,
    e INTEGER,
    PRIMARY KEY (d),
    FOREIGN KEY (a) references R)
```

Which one of the following statements is **CORRECT**?

A. S1 is TRUE and S2 is FALSE
B. Both S1 and S2 are TRUE
C. S1 is FALSE and S2 is TRUE
D. Both S1 and S2 are FALSE

gatecse-2014-set1 databases normal sql

Answer key

5.18.32 SQL: GATE CSE 2014 Set 1 | Question: 54

Given the following schema:

employees(emp-id, first-name, last-name, hire-date, dept-id, salary)

departments(dept-id, dept-name, manager-id, location-id)

You want to display the last names and hire dates of all latest hires in their respective departments in the location $\text{ID } 1700$. You issue the following query:

```
SQL>SELECT last-name, hire-date
   FROM employees
   WHERE (dept-id, hire-date) IN
   (SELECT dept-id, MAX(hire-date)
   FROM employees JOIN departments USING(dept-id)
   WHERE location-id =1700
   GROUP BY dept-id);
```

What is the outcome?

A. It executes but does not give the correct result
B. It executes and gives the correct result.
C. It generates an error because of pairwise comparison.
D. It generates an error because of the GROUP BY clause cannot be used with table joins in a sub-query.

gatecse-2014-set1 databases sql normal

Answer key

5.18.33 SQL: GATE CSE 2014 Set 2 | Question: 54

SQL allows duplicate tuples in relations, and correspondingly defines the multiplicity of tuples in the result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

```
select * from R where a in (select S.a from S)
```

A. select R.* from R, S where R.a=S.a
B. select distinct R.* from R,S where R.a=S.a
C. select R.* from R,(select distinct a from S) as S1 where R.a=S1.a
D. select R.* from R,S where R.a=S.a and is unique R

gatecse-2014-set2 databases sql normal

Answer key

5.18.34 SQL: GATE CSE 2014 Set 3 | Question: 54

Consider the following relational schema:

employee (empId,empName,empDept)

customer (custId,custName,salesRepId,rating)

salesRepId is a foreign key referring to **empId** of the employee relation. Assume that each employee makes a sale to at least one customer. What does the following query return?

```
SELECT empName   FROM employee E
WHERE NOT EXISTS (SELECT custId
   FROM customer C
   WHERE C.salesRepId = E.empId
   AND C.rating <> 'GOOD');
```

A. Names of all the employees with at least one of their customers having a ‘GOOD’ rating.
B. Names of all the employees with at most one of their customers having a 'GOOD' rating.
C. Names of all the employees with none of their customers having a 'GOOD' rating.
D. Names of all the employees with all their customers having a 'GOOD' rating.

gatecse-2014-set3   databases   sql   easy

Answer key

5.18.35 SQL: GATE CSE 2015 Set 1 | Question: 27

Consider the following relation:

Student

Roll_No	Student_Name
1	Raj
2	Rohit
3	Raj

Performance

Roll_No	Course	Marks
1	Math	80
1	English	70
2	Math	75
3	English	80
2	Physics	65
3	Math	80

Consider the following SQL query.

```
SELECT S.Student_Name, Sum(P. Marks)
FROM Student S, Performance P
WHERE S.Roll_No= P.Roll_No
GROUP BY S.STUDENT_Name
```

The numbers of rows that will be returned by the SQL query is________________.

gatecse-2015-set1   databases   sql   normal   numerical-answers

Answer key

5.18.36 SQL: GATE CSE 2015 Set 3 | Question: 3

Consider the following relation

Cinema($theater, address, capacity$)

Which of the following options will be needed at the end of the SQL query

```
SELECT P1.address
FROM Cinema P1
```

such that it always finds the addresses of theaters with maximum capacity?

A. `WHERE P1.capacity >= All (select P2.capacity from Cinema P2)`

B. `WHERE P1.capacity >= Any (select P2.capacity from Cinema P2)`

C. `WHERE P1.capacity > All (select max(P2.capacity) from Cinema P2)`

D.
```
WHERE P1.capacity > Any (select max(P2.capacity) from Cinema P2)
```

gatecse-2015-set3 databases sql normal

Answer key

5.18.37 SQL: GATE CSE 2016 Set 2 | Question: 52

Consider the following database table named water_schemes:

Water_schemes

scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _________.

```
with total (name, capacity) as
   select district_name, sum (capacity)
   from water_schemes
   group by district_name
with total_avg (capacity) as
  select avg (capacity)
  from total
select name
  from total, total_avg
  where total.capacity ≥ total_avg.capacity
```

gatecse-2016-set2 databases sql normal numerical-answers

Answer key

5.18.38 SQL: GATE CSE 2017 Set 1 | Question: 23

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below:

EMP

EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
```

```
(SELECT DeptName, COUNT(EmpId) AS
            EC(DeptName, Num)
FROM EMP
GROUP BY DeptName)
```

The output of executing the SQL query is _____________ .

gatecse-2017-set1 databases sql numerical-answers

Answer key

5.18.39 SQL: GATE CSE 2017 Set 2 | Question: 46

Consider the following database table named top_scorer.

top_scorer

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Muller	Germany	14
Fontaine	France	13
Pele	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Muller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals >ALL (SELECT tb.goals
  FROM top_scorer AS tb
  WHERE tb.country = 'Spain')
AND ta.goals > ANY (SELECT tc.goals
  FROM top_scorer AS tc
  WHERE tc.country='Germany')
```

The number of tuples returned by the above SQL query is ______

gatecse-2017-set2 databases sql numerical-answers

Answer key

5.18.40 SQL: GATE CSE 2018 | Question: 12

Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock(isbn, copies)

Query 1:

```
SELECT B.isbn, S.copies FROM Book B INNER JOIN Stock S ON B.isbn=S.isbn;
```

Query 2:

```
SELECT B.isbn, S.copies FROM Book B LEFT OUTER JOIN Stock S ON B.isbn=S.isbn;
```

Query 3:

```
SELECT B.isbn, S,copies FROM Book B RIGHT OUTER JOIN Stock S ON B.isbn=S.isbn
```

Query 4:

```
SELECT B.isbn, S.copies FROM Book B FULL OUTER JOIN Stock S ON B.isbn=S.isbn
```

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

A. Query 1 B. Query 2 C. Query 3 D. Query 4

gatecse-2018 databases sql easy one-mark

Answer key

5.18.41 SQL: GATE CSE 2019 | Question: 51

A relational database contains two tables Student and Performance as shown below:

Table: student

Roll_no	Student_name
1	Amit
2	Priya
3	Vinit
4	Rohan
5	Smita

Table: Performance

Roll_no	Subject_code	Marks
1	A	86
1	B	95
1	C	90
2	A	89
2	C	92
3	C	80

The primary key of the Student table is Roll_no. For the performance table, the columns Roll_no. and Subject_code together form the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum(P.Marks)
FROM Student S, Performance P
WHERE P.Marks >84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _________

gatecse-2019 numerical-answers databases sql two-marks

Answer key

5.18.42 SQL: GATE CSE 2020 | Question: 13

Consider a relational database containing the following schemas.

Catalogue

sno	pno	cost
$S1$	$P1$	150
$S1$	$P2$	50
$S1$	$P3$	100
$S2$	$P4$	200
$S2$	$P5$	250
$S3$	$P1$	250
$S3$	$P2$	150
$S3$	$P5$	300
$S3$	$P4$	250

Suppliers

sno	sname	location
$S1$	M/s Royal furniture	Delhi
$S2$	M/s Balaji furniture	Bangalore
$S3$	M/s Premium furniture	Chennai

Parts

pno	pname	part_spec
$P1$	Table	Wood
$P2$	Chair	Wood
$P3$	Table	Steel
$P4$	Almirah	Steel
$P5$	Almirah	Wood

The primary key of each table is indicated by underlining the constituent fields.

```
SELECT s.sno, s.sname
FROM Suppliers s, Catalogue c
WHERE s.sno=c.sno AND
  cost > (SELECT AVG (cost)
```

```
FROM Catalogue
WHERE pno = 'P4'
GROUP BY pno) ;
```

The number of rows returned by the above SQL query is

A. 4 B. 5 C. 0 D. 2

gatecse-2020 databases sql one-mark

Answer key

5.18.43 SQL: GATE CSE 2021 Set 1 | Question: 23

A relation $r(A, B)$ in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed.

The estimated number of tuples in the output of $\sigma_{(A>10) \vee (B=18)}(r)$ is ____________.

gatecse-2021-set1 databases sql numerical-answers one-mark

Answer key

5.18.44 SQL: GATE CSE 2021 Set 2 | Question: 31

The relation scheme given below is used to store information about the employees of a company, where **empId** is the key and **deptId** indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.

$$\text{emp}(\underline{\text{empId}}, \text{name}, \text{gender}, \text{salary}, \text{deptId})$$

Consider the following SQL query:

```
select deptId, count(*)
from emp
where gender = "female" and salary > (select avg(salary)from emp)
group by deptId;
```

The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of

A. employees in the department
B. employees in the company
C. female employees in the department
D. female employees in the company

gatecse-2021-set2 databases sql easy two-marks

Answer key

5.18.45 SQL: GATE CSE 2022 | Question: 46

Consider the relational database with the following four schemas and their respective instances.

- Student(sNo, sName, dNo) Dept(dNo, dName)
- Course(cNo, cName, dNo) Register(sNo, cNo)

	Students	
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

	Depth
dNo	dName
D01	CSE
D02	EEE

	Course	
cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

	Register
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

SQL query

```
SELECT * FROM Student AS S WHERE NOT EXIST

 (SELECT cNo FROM Course WHERE dNo = “D01”

     EXCEPT

  SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is ____________.

gatecse-2022 numerical-answers databases sql two-marks

Answer key

5.18.46 SQL: GATE CSE 2023 | Question: 51

Consider the following table named Student in a relational database. The primary key of this table is rollNum.

Student

rollNum	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

The SQL query below is executed on this database.

```
SELECT *
FROM Student
WHERE gender = 'F' AND
  marks > 65;
```

The number of rows returned by the query is ___________.

gatecse-2023 databases sql numerical-answers two-marks easy

Answer key

5.18.47 SQL: GATE CSE 2025 | Set 1 | Question: 45

Consider the following database tables of a sports league.

```
player (pid, pname, age)        team(tid, tname, city, cid)
coach (cid, cname)             members (pid,tid)
```

An instance of the table and an SQL query are given.

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN (P.age)
FROM player P
WHERE P.pid IN (
  SELECT M.pid
  FROM team T, coach C, members M
  WHERE C.cname = 'Mark'
     AND T.cid = C.cid
     AND M.tid = T.tid
  )
```

The value returned by the given SQL query is __________. (Answer in integer)

gatecse2025-set1 databases sql numerical-answers easy two-marks

Answer key

5.18.48 SQL: GATE CSE 2025 | Set 2 | Question: 44

Consider the following relational schema:

Students (<u>rollno: integer</u> , name: string, age: integer, cgpa: real)

Courses (<u>courseno: integer</u> , cname: string, credits: integer)

Enrolled (<u>rollno: integer</u> , <u>courseno: integer</u> , grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

```
A. SELECT S.name
  FROM Students S
  WHERE EXISTS (SELECT * FROM Enrolled E
          WHERE E.courseno = 1470
               AND E.rollno = S.rollno);
```

```
B. SELECT S.name
  FROM Students S
  WHERE SIZEOF (SELECT * FROM Enrolled E
          WHERE E.courseno = 1470
               AND E.rollno = S.rollno) > 0;
```

```
C. SELECT S.name
  FROM Students S
  WHERE 0 < (SELECT COUNT(*)
        FROM Enrolled E
        WHERE E.courseno = 1470
            AND E.rollno = S.rollno);
```

```
D. SELECT S.name
  FROM Students S NATURAL JOIN Enrolled E
  WHERE E.courseno = 1470;
```

gatecse2025-set2 databases sql multiple-selects two-marks

Answer key

5.18.49 SQL: GATE DA 2025 | Question: 23

On a relation named Loan of a bank:

Loan		
loan_number	branch_name	amount
L11	Banjara Hills	90000
L14	Kondapur	50000
L15	SR Nagar	40000
L22	SR Nagar	25000
L23	Balanagar	80000
L25	Kondapur	70000
L19	SR Nagar	65000

the following SQL query is executed.

```
SELECT L1.loan_number
FROM Loan L1
WHERE L1.amount > (SELECT MAX (L2.amount)
            FROM Loan L2
            WHERE L2.branch_name = 'SR Nagar');
```

The number of rows returned by the query is __________ (Answer in integer).

gateda-2025 databases sql numerical-answers one-mark

Answer key

5.18.50 SQL: GATE DS&AI 2024 | Question: 21

Consider the following two tables named Raider and Team in a relational database maintained by a Kabaddi league. The attribute ID in table Team references the primary key of the Raider table, ID.

Raider			
ID	Name	Raids	Raidpoints
1	Arjun	200	250
2	Ankush	190	219
3	Sunil	150	200
4	Reza	150	190
5	Pratham	175	220
6	Gopal	193	215

Team		
City	ID	BidPoints
Jaipur	2	200
Patna	3	195
Hyderabad	5	175
Jaipur	1	250
Patna	4	200
Jaipur	6	200

The SQL query described below is executed on this database:

```
SELECT *
FROM Raider, Team
WHERE Raider.ID=Team.ID AND City="Jaipur" AND
RaidPoints > 200;
```

The number of rows returned by this query is _________.

gate-ds-ai-2024 numerical-answers databases sql one-mark

Answer key

5.18.51 SQL: GATE DS&AI 2024 | Question: 45

An OTT company is maintaining a large disk-based relational database of different movies with the following schema:

$$\text{Movie (ID, CustomerRating)}$$
$$\text{Genre (ID, Name)}$$
$$\text{Movie\_Genre (MovieID, GenreID)}$$

Consider the following SQL query on the relation database above:

```
SELECT *
FROM Movie, Genre, Movie_Genre
WHERE
   Movie.CustomerRating > 3.4 AND
   Genre.Name = "Comedy" AND
   Movie_Genre.MovieID = Movie.ID AND
   Movie_Genre.GenreID = Genre.ID;
```

This SQL query can be sped up using which of the following indexing options?

A. B^{+}tree on all the attributes.
B. Hash index on Genre.Name and B^{+}tree on the remaining attributes.
C. Hash index on Movie.CustomerRating and B^{+}tree on the remaining attributes.
D. Hash index on all the attributes.

gate-ds-ai-2024 sql databases multiple-selects two-marks

Answer key

5.18.52 SQL: GATE IT 2004 | Question: 74

A relational database contains two tables student and department in which student table has columns roll_no, name and dept_id and department table has columns dept_id and dept_name. The following insert statements were executed successfully to populate the empty tables:

```
Insert into department values (1, 'Mathematics')
Insert into department values (2, 'Physics')
Insert into student values (l, 'Navin', 1)
Insert into student values (2, 'Mukesh', 2)
Insert into student values (3, 'Gita', 1)
```

How many rows and columns will be retrieved by the following SQL statement?

```
Select * from student, department
```

A. 0 row and 4 columns
B. 3 rows and 4 columns
C. 3 rows and 5 columns
D. 6 rows and 5 columns

gateit-2004 databases sql normal

Answer key

5.18.53 SQL: GATE IT 2004 | Question: 76

A table T1 in a relational database has the following rows and columns:

Roll no.	Marks
1	10
2	20
3	30
4	NULL

The following sequence of SQL statements was successfully executed on table T1.

```
Update T1 set marks = marks + 5
Select avg(marks) from T1
```

What is the output of the select statement?

A. 18.75
B. 20
C. 25
D. Null

gateit-2004 databases sql normal

Answer key

5.18.54 SQL: GATE IT 2004 | Question: 78

Consider two tables in a relational database with columns and rows as follows:

Table: Student

Roll_no	Name	Dept_id
1	ABC	1
2	DEF	1
3	GHI	2
4	JKL	3

Table: Department

Dept_id	Dept_name
1	A
2	B
3	C

Roll_no is the primary key of the Student table, Dept_id is the primary key of the Department table and Student.Dept_id is a foreign key from Department.Dept_id
What will happen if we try to execute the following two SQL statements?

i. update Student set Dept_id = Null where Roll_on = 1
ii. update Department set Dept_id = Null where Dept_id = 1

A. Both i and ii will fail
B. i will fail but ii will succeed
C. i will succeed but ii will fail
D. Both i and ii will succeed

gateit-2004 databases sql normal

Answer key

5.18.55 SQL: GATE IT 2005 | Question: 69

In an inventory management system implemented at a trading corporation, there are several tables designed to hold all the information. Amongst these, the following two tables hold information on which items are supplied by which suppliers, and which warehouse keeps which items along with the stock-level of these items.

Supply = (supplierid, itemcode)
Inventory = (itemcode, warehouse, stocklevel)

For a specific information required by the management, following SQL query has been written

```
Select distinct STMP.supplierid
From Supply as STMP
Where not unique (Select ITMP.supplierid
                    From Inventory, Supply as ITMP
                    Where STMP.supplierid = ITMP.supplierid
                    And ITMP.itemcode = Inventory.itemcode
                    And Inventory.warehouse = 'Nagpur');
```

For the warehouse at Nagpur, this query will find all suppliers who

A. do not supply any item
B. supply exactly one item
C. supply one or more items
D. supply two or more items

gateit-2005 databases sql normal

Answer key

5.18.56 SQL: GATE IT 2006 | Question: 84

Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are *did* and *cid* respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
29	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
85	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	Cid	day
22	101	10 / 10 / 06
22	102	10 / 10 / 06
22	103	08 / 10 / 06
22	104	07 / 10 / 06
31	102	10 / 11 / 16
31	103	06 / 11 / 16
31	104	12 / 11 / 16
64	101	05 / 09 / 06
64	102	08 / 09 / 06
74	103	08 / 09 / 06

C: Cars relation

Cid	Cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

What is the output of the following SQL query?

```
select D.dname
from Drivers D
where D.did in  (
                select R.did
                from Cars C, Reserves R
                where R.cid = C.cid and C.colour = 'red'
                intersect
                select R.did
                from Cars C, Reserves R
                where R.cid  = C.cid and C.colour = 'green'
                 )
```

A. Karthikeyan, Boris
B. Sachin, Salman
C. Karthikeyan, Boris, Sachin
D. Schumacher, Senna

gateit-2006 databases sql normal

Answer key

5.18.57 SQL: GATE IT 2006 | Question: 85

Consider a database with three relation instances shown below. The primary keys for the Drivers and Cars relation are *did* and *cid* respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers relation

did	dname	rating	age
22	Karthikeyan	7	25
29	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
85	Rahul	3	25
95	Ralph	3	53

R: Reserves relation

did	Cid	day
22	101	10 – 10 – 06
22	102	10 – 10 – 06
22	103	08 – 10 – 06
22	104	07 – 10 – 06
31	102	10 – 11 – 16
31	103	06 – 11 – 16
31	104	12 – 11 – 16
64	101	05 – 09 – 06
64	102	08 – 09 – 06
74	103	08 – 09 – 06

C: Cars relation

Cid	Cname	colour
101	Renault	blue
102	Renault	red
103	Ferrari	green
104	Jaguar	red

```
select D.dname
from Drivers D
where D.did in  (
                 select R.did
                 from Cars C, Reserves R
                 where R.cid = C.cid and C.colour = 'red'
                 intersect
                 select R.did
                 from Cars C, Reserves R
                 where R.cid  = C.cid and C.colour = 'green'
                  )
```

Let n be the number of comparisons performed when the above SQL query is optimally executed. If linear search is used to locate a tuple in a relation using primary key, then n lies in the range:

A. $36 - 40$ B. $44 - 48$ C. $60 - 64$ D. $100 - 104$

gateit-2006 databases sql normal

Answer key

5.18.58 SQL: GATE IT 2008 | Question: 74

Consider the following relational schema:

- Student($\underline{\text{school-id, sch-roll-no}}$, sname, saddress)
- School($\underline{\text{school-id}}$, sch-name, sch-address, sch-phone)
- Enrolment($\underline{\text{school-id, sch-roll-no}}$, erollno, examname)
- ExamResult($\underline{\text{erollno, examname}}$, marks)

What does the following SQL query output?

```
SELECT sch-name, COUNT (*)
FROM School C, Enrolment E, ExamResult R
WHERE E.school-id = C.school-id
AND
E.examname = R.examname AND E.erollno = R.erollno
AND
R.marks = 100 AND E.school-id IN (SELECT school-id
                                  FROM student
                                  GROUP BY school-id
                                   HAVING COUNT (*) > 200)
GROUP By school-id
```

A. for each school with more than 200 students appearing in exams, the name of the school and the number of $100s$ scored by its students

B. for each school with more than 200 students in it, the name of the school and the number of $100s$ scored by its students

C. for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam

D. nothing; the query has a syntax error

gateit-2008 databases sql normal

Answer key

5.19 Transaction and Concurrency (30)

5.19.1 Transaction and Concurrency: GATE CSE 1999 | Question: 2.6

For the schedule given below, which of the following is correct:

1	Read A	
2		Read B
3	Write A	
4		Read A
5		Write A
6		Write B
7	Read B	
8	Write B	

A. This schedule is serializable and can occur in a scheme using 2PL protocol

B. This schedule is serializable but cannot occur in a scheme using 2PL protocol

C. This schedule is not serializable but can occur in a scheme using 2PL protocol

D. This schedule is not serializable and cannot occur in a scheme using 2PL protocol

gate1999 databases transaction-and-concurrency normal

Answer key

5.19.2 Transaction and Concurrency: GATE CSE 2003 | Question: 29, ISRO2009-73

Which of the following scenarios may lead to an irrecoverable error in a database system?

A. A transaction writes a data item after it is read by an uncommitted transaction
B. A transaction reads a data item after it is read by an uncommitted transaction
C. A transaction reads a data item after it is written by a committed transaction
D. A transaction reads a data item after it is written by an uncommitted transaction

gatecse-2003 databases transaction-and-concurrency easy isro2009

Answer key

5.19.3 Transaction and Concurrency: GATE CSE 2003 | Question: 87

Consider three data items $D1, D2,$ and $D3,$ and the following execution schedule of transactions $T1, T2,$ and $T3.$ In the diagram, $R(D)$ and $W(D)$ denote the actions reading and writing the data item D respectively.

T1	T2	T3
	R(D3);	
	R(D2);	
	W(D2);	
		R(D2);
		R(D3);
R(D1);		
W(D1);		
		W(D2);
		W(D3);
	R(D1);	
R(D2);		
W(D2);		
	W(D1);	

Which of the following statements is correct?

A. The schedule is serializable as $T2; T3; T1$
B. The schedule is serializable as $T2; T1; T3$
C. The schedule is serializable as $T3; T2; T1$
D. The schedule is not serializable

gatecse-2003 databases transaction-and-concurrency normal

Answer key

5.19.4 Transaction and Concurrency: GATE CSE 2006 | Question: 20, ISRO2015-17

Consider the following log sequence of two transactions on a bank account, with initial balance $12000,$ that transfer 2000 to a mortgage payment and then apply a 5% interest.

1. T1 start
2. T1 B old $= 12000$ new $= 10000$
3. T1 M old $= 0$ new $= 2000$
4. T1 commit
5. T2 start
6. T2 B old $= 10000$ new $= 10500$
7. T2 commit

Suppose the database system crashes just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

A. We must redo log record 6 to set B to 10500
B. We must undo log record 6 to set B to 10000 and then redo log records 2 and 3
C. We need not redo log records 2 and 3 because transaction T1 has committed
D. We can apply redo and undo operations in arbitrary order because they are idempotent

gatecse-2006 databases transaction-and-concurrency normal isro2015

Answer key

5.19.5 Transaction and Concurrency: GATE CSE 2007 | Question: 64

Consider the following schedules involving two transactions. Which one of the following statements is **TRUE**?

- $S_1 : r_1(X); r_1(Y); r_2(X); r_2(Y); w_2(Y); w_1(X)$
- $S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$

A. Both S_1 and S_2 are conflict serializable.

B. S_1 is conflict serializable and S_2 is not conflict serializable.

C. S_1 is not conflict serializable and S_2 is conflict serializable.

D. Both S_1 and S_2 are not conflict serializable.

gatecse-2007 databases transaction-and-concurrency normal

Answer key

5.19.6 Transaction and Concurrency: GATE CSE 2009 | Question: 43

Consider two transactions T_1 and T_2, and four schedules S_1, S_2, S_3, S_4, of T_1 and T_2 as given below:

$T_1 : R_1[x]W_1[x]W_1[y]$

$T_2 : R_2[x]R_2[y]W_2[y]$

$S_1 : R_1[x]R_2[x]R_2[y]W_1[x]W_1[y]W_2[y]$

$S_2 : R_1[x]R_2[x]R_2[y]W_1[x]W_2[y]W_1[y]$

$S_3 : R_1[x]W_1[x]R_2[x]W_1[y]R_2[y]W_2[y]$

$S_4 : R_2[x]R_2[y]R_1[x]W_1[x]W_1[y]W_2[y]$

Which of the above schedules are conflict-serializable?

A. S_1 and S_2

B. S_2 and S_3

C. S_3 only

D. S_4 only

gatecse-2009 databases transaction-and-concurrency normal

Answer key

5.19.7 Transaction and Concurrency: GATE CSE 2010 | Question: 20

Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?

I. 2-phase locking

II. Time-stamp ordering

A. I only

B. II only

C. Both I and II

D. Neither I nor II

gatecse-2010 databases transaction-and-concurrency normal

Answer key

5.19.8 Transaction and Concurrency: GATE CSE 2010 | Question: 42

Consider the following schedule for transactions $T1, T2$ and $T3$:

T1	T2	T3
Read(X)		
	Read(Y)	
		Read(Y)
	Write(Y)	
Write(X)		
		Write(X)
	Read(X)	
	Write(X)	

Which one of the schedules below is the correct serialization of the above?

A. $T1 \rightarrow T3 \rightarrow T2$
B. $T2 \rightarrow T1 \rightarrow T3$
C. $T2 \rightarrow T3 \rightarrow T1$
D. $T3 \rightarrow T1 \rightarrow T2$

gatecse-2010 databases transaction-and-concurrency normal

Answer key

5.19.9 Transaction and Concurrency: GATE CSE 2012 | Question: 27

Consider the following transactions with data items P and Q initialized to zero:

T_1	read (P); read (Q); if $\mathrm{P} = 0$ then $\mathrm{Q} := \mathrm{Q} + 1$; write (Q)
T_2	read (Q); read (P); if $\mathrm{Q} = 0$ then $\mathrm{P} := \mathrm{P} + 1$; write (P)

Any non-serial interleaving of **T1** and **T2** for concurrent execution leads to

A. a serializable schedule
B. a schedule that is not conflict serializable
C. a conflict serializable schedule
D. a schedule for which a precedence graph cannot be drawn

gatecse-2012 databases transaction-and-concurrency normal

Answer key

5.19.10 Transaction and Concurrency: GATE CSE 2015 Set 2 | Question: 1

Consider the following transaction involving two bank accounts x and y.

```
read(x); x:=x-50; write (x); read(y); y:=y+50; write(y)
```

The constraint that the sum of the accounts x and y should remain constant is that of

A. Atomicity
B. Consistency
C. Isolation
D. Durability

gatecse-2015-set2 databases transaction-and-concurrency easy

Answer key

5.19.11 Transaction and Concurrency: GATE CSE 2015 Set 2 | Question: 46

Consider a simple checkpointing protocol and the following set of operations in the log.

(start, T4); (write, T4, y, 2, 3); (start, T1); (commit, T4); (write, T1, z, 5, 7);

(checkpoint);

(start, T2); (write, T2, x, 1, 9); (commit, T2); (start, T3); (write, T3, z, 7, 2);

If a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

A. Undo: T3, T1; Redo: T2
B. Undo: T3, T1; Redo: T2, T4
C. Undo: none; Redo: T2, T4, T3, T1
D. Undo: T3, T1, T4; Redo: T2

gatecse-2015-set2 databases transaction-and-concurrency normal

Answer key

5.19.12 Transaction and Concurrency: GATE CSE 2015 Set 3 | Question: 29

Consider the partial Schedule S involving two transactions $T1$ and $T2$. Only the $read$ and the $write$ operations have been shown. The $read$ operation on data item P is denoted by $read(P)$ and $write$

operation on data item P is denoted by $write(P)$.

Schedule S

Time Instance	Transaction ID **T1**	**T2**
1	read(A)	
2	write(A)	
3		read(C)
4		write(C)
5		read(B)
6		write(B)
7		read(A)
8		commit
9	read(B)	

Suppose that the transaction $T1$ fails immediately after time instance 9. Which of the following statements is correct?

A. $T2$ must be aborted and then both $T1$ and $T2$ must be re-started to ensure transaction atomicity
B. Schedule S is non-recoverable and cannot ensure transaction atomicity
C. Only $T2$ must be aborted and then re-started to ensure transaction atomicity
D. Schedule S is recoverable and can ensure transaction atomicity and nothing else needs to be done

gatecse-2015-set3 databases transaction-and-concurrency normal

Answer key

5.19.13 Transaction and Concurrency: GATE CSE 2016 Set 1 | Question: 22

Which one of the following is NOT a part of the ACID properties of database transactions?

A. Atomicity B. Consistency C. Isolation D. Deadlock-freedom

gatecse-2016-set1 databases transaction-and-concurrency easy

Answer key

5.19.14 Transaction and Concurrency: GATE CSE 2016 Set 1 | Question: 51

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \dots, O_k\}$. This is done in the following manner:

- Step 1. T acquires exclusive locks to $O_1, \dots, O_k$ in increasing order of their addresses.
- Step 2. The required operations are performed .
- Step 3. All locks are released

This protocol will

A. guarantee serializability and deadlock-freedom
B. guarantee neither serializability nor deadlock-freedom
C. guarantee serializability but not deadlock-freedom
D. guarantee deadlock-freedom but not serializability.

gatecse-2016-set1 databases transaction-and-concurrency normal

Answer key

5.19.15 Transaction and Concurrency: GATE CSE 2016 Set 2 | Question: 22

Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S

with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

A. Topological order
B. Depth-first order
C. Breadth-first order
D. Ascending order of the transaction indices

gatecse-2016-set2 databases transaction-and-concurrency normal

Answer key

5.19.16 Transaction and Concurrency: GATE CSE 2016 Set 2 | Question: 51

Consider the following database schedule with two transactions T_1 and T_2.

$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$

Where $r_i(Z)$ denotes a read operation by transaction T_i on a variable Z, $w_i(Z)$ denotes a write operation by T_i on a variable Z and a_i denotes an abort by transaction T_i.

Which one of the following statements about the above schedule is **TRUE**?

A. S is non-recoverable.
B. S is recoverable, but has a cascading abort.
C. S does not have a cascading abort.
D. S is strict.

gatecse-2016-set2 databases transaction-and-concurrency normal

Answer key

5.19.17 Transaction and Concurrency: GATE CSE 2017 Set 1 | Question: 42

In a database system, unique timestamps are assigned to each transaction using Lamport's logical clock. Let $TS(T_1)$ and $TS(T_2)$ be the timestamps of transactions T_1 and T_2 respectively. Besides, T_1 holds a lock on the resource R, and T_2 has requested a conflicting lock on the same resource R. The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.

if $TS(T_2) < TS(T_1)$ then

T_1 is killed

else T_2 waits.

Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?

A. The database system is both deadlock-free and starvation-free.
B. The database system is deadlock-free, but not starvation-free.
C. The database system is starvation-free, but not deadlock-free.
D. The database system is neither deadlock-free nor starvation-free.

gatecse-2017-set1 databases timestamp-ordering normal transaction-and-concurrency

Answer key

5.19.18 Transaction and Concurrency: GATE CSE 2019 | Question: 11

Consider the following two statements about database transaction schedules:

I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.
II. Timestamp-ordering concurrency control protocol with Thomas' Write Rule can generate view serializable schedules that are not conflict serializable

Which of the above statements is/are TRUE?

A. I only
B. II only
C. Both I and II
D. Neither I nor II

gatecse-2019 databases transaction-and-concurrency one-mark

Answer key

5.19.19 Transaction and Concurrency: GATE CSE 2020 | Question: 37

Consider a schedule of transactions T_1 and T_2:

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for "Read(X)" and WX stands for "Write(X)". Which one of the following schedules is conflict equivalent to the above schedule?

A.

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit

B.

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit

C.

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit

D.

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

gatecse-2020 databases transaction-and-concurrency two-marks

Answer key

5.19.20 Transaction and Concurrency: GATE CSE 2021 Set 1 | Question: 13

Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery.

Which of the following statements is/are correct?

A. The same undo and redo list will be used while recovering again
B. The system cannot recover any further
C. All the transactions that are already undone and redone will not be recovered again
D. The database will become inconsistent

gatecse-2021-set1 multiple-selects databases transaction-and-concurrency one-mark

Answer key

5.19.21 Transaction and Concurrency: GATE CSE 2024 | Set 2 | Question: 17

Which of the following statements about the Two Phase Locking ($2PL$) protocol is/are TRUE?

A. $2PL$ permits only serializable schedules
B. With $2PL$, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
C. With $2PL$, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
D. A deadlock is possible with $2PL$

gatecse2024-set2 databases two-phase-locking-protocol multiple-selects transaction-and-concurrency one-mark

Answer key

5.19.22 Transaction and Concurrency: GATE CSE 2024 | Set 2 | Question: 9

Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

A. durability
B. atomicity
C. consistency
D. isolation

gatecse2024-set2 databases transaction-and-concurrency one-mark

Answer key☟

5.19.23 Transaction and Concurrency: GATE CSE 2025 | Set 1 | Question: 5

A schedule of three database transactions $T_1, T_2,$ and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1, 2, 3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?

$$R_1(X)W_1(Y)R_2(X)R_2(Y)R_3(Y)\,\text{ABORT}(T_1)$$

A. Only T_2
B. Only T_3
C. Both T_2 and T_3
D. Neither T_2 nor T_3

gatecse2025-set1 databases transaction-and-concurrency one-mark

Answer key☟

5.19.24 Transaction and Concurrency: GATE CSE 2025 | Set 2 | Question: 17

An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs.11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000.

Which of the following properties is/are certain to have been violated by the system?

A. Atomicity
B. Consistency
C. Isolation
D. Durability

gatecse2025-set2 databases transaction-and-concurrency multiple-selects one-mark

Answer key☟

5.19.25 Transaction and Concurrency: GATE IT 2004 | Question: 21

Which level of locking provides the highest degree of concurrency in a relational database ?

A. Page
B. Table
C. Row
D. Page, table and row level locking allow the same degree of concurrency

gateit-2004 databases normal transaction-and-concurrency

Answer key☟

5.19.26 Transaction and Concurrency: GATE IT 2004 | Question: 77

Consider the following schedule S of transactions $T1$ and $T2$:

T1	T2
$\text{Read}(A)$	
$A = A - 10$	
	$\text{Read}(A)$
	$\text{Temp} = 0.2{*}A$
	$\text{Write}(A)$
	$\text{Read}(B)$
$\text{Write}(A)$	
$\text{Read}(B)$	
$B = B + 10$	
$\text{Write}(B)$	
	$B = B + \text{Temp}$
	$\text{Write}(B)$

Which of the following is TRUE about the schedule S ?

A. S is serializable only as $T1, T2$
B. S is serializable only as $T2, T1$
C. S is serializable both as $T1, T2$ and $T2, T1$
D. S is not serializable either as $T1, T2$ or as $T2, T1$

gateit-2004 databases transaction-and-concurrency normal

Answer key☟

5.19.27 Transaction and Concurrency: GATE IT 2005 | Question: 24

Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

A. Except in case of an Operating System crash
B. Except in case of a Disk crash
C. Except in case of a power failure
D. Always, even if there is a failure of any kind

gateit-2005 databases transaction-and-concurrency easy

Answer key☟

5.19.28 Transaction and Concurrency: GATE IT 2005 | Question: 67

A company maintains records of sales made by its salespersons and pays them commission based on each individual's total sales made in a year. This data is maintained in a table with following schema:

salesinfo = (salespersonid, totalsales, commission)

In a certain year, due to better business results, the company decides to further reward its salespersons by enhancing the commission paid to them as per the following formula:

If commission ≤ 50000, enhance it by 2%
If $50000 <$ commission ≤ 100000, enhance it by 4%
If commission > 100000, enhance it by 6%

The IT staff has written three different SQL scripts to calculate enhancement for each slab, each of these scripts is to run as a separate transaction as follows:

T1
```
Update salesinfo
Set commission = commission * 1.02
Where commission < = 50000;
```

T2
```
Update salesinfo
Set commission = commission * 1.04
Where commission > 50000 and
commission is < = 100000;
```

T3
```
Update salesinfo
Set commission = commission * 1.06
Where commission > 100000;
```

Which of the following options of running these transactions will update the commission of all salespersons correctly

A. Execute T1 followed by T2 followed by T3
B. Execute T2, followed by T3; T1 running concurrently throughout
C. Execute T3 followed by T2; T1 running concurrently throughout
D. Execute T3 followed by T2 followed by T1

gateit-2005 databases transaction-and-concurrency normal

Answer key

5.19.29 Transaction and Concurrency: GATE IT 2007 | Question: 66

Consider the following two transactions: $T1$ and $T2$.

$$
\begin{array}{llll}
T1: & \text{read (A);} & T2: & \text{read (B);} \\
 & \text{read (B);} & & \text{read (A);} \\
 & \text{If } A = 0 \text{ then } B \leftarrow B + 1; & & \text{If } B \neq 0 \text{ then } A \leftarrow A - 1; \\
 & \text{write (B);} & & \text{write (A);}
\end{array}
$$

Which of the following schemes, using shared and exclusive locks, satisfy the requirements for strict two phase locking for the above transactions?

A.
$$
\begin{array}{llll}
S1: & \text{lock S(A);} & S2: & \text{lock S(B);} \\
 & \text{read (A);} & & \text{read (B);} \\
 & \text{lock S(B);} & & \text{lock S(A);} \\
 & \text{read (B);} & & \text{read (A);} \\
 & \text{If } A = 0 & & \text{If } B \neq 0 \\
 & \text{then } B \leftarrow B + 1; & & \text{then } A \leftarrow A - 1; \\
 & \text{write (B);} & & \text{write (A);} \\
 & \text{commit;} & & \text{commit;} \\
 & \text{unlock (A);} & & \text{unlock (B);} \\
 & \text{unlock (B);} & & \text{unlock (A);}
\end{array}
$$

B.
$$
\begin{array}{llll}
S1: & \text{lock X(A);} & S2: & \text{lock X(B);} \\
 & \text{read (A);} & & \text{read (B);} \\
 & \text{lock X(B);} & & \text{lock X(A);} \\
 & \text{read (B);} & & \text{read (A);} \\
 & \text{If } A = 0 & & \text{If } B \neq 0 \\
 & \text{then } B \leftarrow B + 1; & & \text{then } A \leftarrow A - 1; \\
 & \text{write (B);} & & \text{write (A);} \\
 & \text{unlock (A);} & & \text{unlock (A);} \\
 & \text{commit;} & & \text{commit;} \\
 & \text{unlock (B);} & & \text{unlock (A);}
\end{array}
$$

C.

$S1:$	lock S(A);	$S2:$	lock S(B);
	read (A);		read (B);
	lock X(B);		lock X(A);
	read (B);		read (A);
	If $A=0$		If $B \neq 0$
	then $B \leftarrow B+1$;		then $A \leftarrow A-1$;
	write (B);		write (A);
	unlock (A);		unlock (B);
	commit;		commit;
	unlock (B);		unlock (A);

D.

$S1:$	lock S(A);	$S2:$	lock S(B);
	read (A);		read (B);
	lock X(B);		lock X(A);
	read (B);		read (A);
	If $A=0$		If $B \neq 0$
	then $B \leftarrow B+1$;		then $A \leftarrow A-1$;
	write (B);		write (A);
	unlock (A);		unlock (A);
	unlock (B);		unlock (A);
	commit;		commit;

gateit-2007 databases transaction-and-concurrency normal

Answer key

5.19.30 Transaction and Concurrency: GATE IT 2008 | Question: 63

Consider the following three schedules of transactions T1, T2 and T3. [Notation: In the following NYO represents the action Y (R for read, W for write) performed by transaction N on object O.]

$(S1)$	$2RA$	$2WA$	$3RC$	$2WB$	$3WA$	$3WC$	$1RA$	$1RB$	$1WA$	$1WB$
$(S2)$	$3RC$	$2RA$	$2WA$	$2WB$	$3WA$	$1RA$	$1RB$	$1WA$	$1WB$	$3WC$
$(S3)$	$2RA$	$3RC$	$3WA$	$2WA$	$2WB$	$3WC$	$1RA$	$1RB$	$1WA$	$1WB$

Which of the following statements is TRUE?

A. S1, S2 and S3 are all conflict equivalent to each other
B. No two of S1, S2 and S3 are conflict equivalent to each other
C. S2 is conflict equivalent to S3, but not to S1
D. S1 is conflict equivalent to S2, but not to S3

gateit-2008 databases transaction-and-concurrency normal

Answer key

5.20 Tuple Relational Calculus (1)

5.20.1 Tuple Relational Calculus: GATE CSE 2025 | Set 1 | Question: 29

Consider two relations describing teams and players in a sports league:

- teams(tid, tname): tid, tname are team-id and team-name, respectively
- players(pid,pname,tid): pid, pname, and tid denote player-id, playername and the team-id of the player,

respectively

Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having tname as ' MI' ?

A. $\{p.\text{ pname } \mid p \in \text{ players } \wedge \exists t\,(t \in \text{ teams } \wedge p.tid = t.tid \wedge t.tname =' MI')\}$
B. $\{p.\text{ pname } \mid p \in \text{ teams } \wedge \exists t\,(t \in \text{ players } \wedge p.tid = t.tid \wedge t.tname =' MI')\}$
C. $\{p.\text{ pname } \mid p \in \text{ players } \wedge \exists t\,(t \in \text{ teams } \wedge t.\text{ tname } =' MI')\}$
D. $\{p.\text{ pname } \mid p \in \text{ teams } \wedge \exists t\,(t \in \text{ players } \wedge t.\text{ tname } =' MI')\}$

gatecse2025-set1 databases tuple-relational-calculus two-marks

Answer key☟

Answer Keys

5.1.1	N/A	5.1.2	N/A	5.1.3	N/A	5.1.4	N/A	5.1.5	B
5.1.6	N/A	5.1.7	B	5.1.8	N/A	5.1.9	N/A	5.1.10	N/A
5.1.11	C	5.1.12	B	5.1.13	C	5.1.14	D	5.1.15	A
5.1.16	C	5.1.17	C	5.1.18	B	5.1.19	5	5.1.20	50
5.1.21	A	5.1.22	52	5.1.23	B	5.1.24	A	5.1.25	B;D
5.1.26	33:33	5.1.27	C	5.1.28	A	5.1.29	C	5.1.30	A
5.1.31	A	5.2.1	N/A	5.2.2	A	5.2.3	8	5.2.4	19
5.2.5	B	5.3.1	D	5.3.2	C	5.3.3	A	5.3.4	54
5.3.5	B	5.3.6	B	5.3.7	A	5.3.8	B;C;D	5.3.9	B
5.4.1	False	5.5.1	True	5.5.2	N/A	5.5.3	N/A	5.5.4	N/A
5.5.5	N/A	5.5.6	N/A	5.5.7	A;B;D	5.5.8	False	5.5.9	N/A
5.5.10	A	5.5.11	D	5.5.12	N/A	5.5.13	B	5.5.14	D
5.5.15	B	5.5.16	C	5.5.17	A	5.5.18	N/A	5.5.19	C
5.5.20	C	5.5.21	D	5.5.22	B	5.5.23	C	5.5.24	D
5.5.25	C	5.5.26	D	5.5.27	C	5.5.28	C	5.5.29	B
5.5.30	A	5.5.31	B	5.5.32	A	5.5.33	C	5.5.34	B
5.5.35	B	5.5.36	A	5.5.37	B	5.5.38	C	5.5.39	A
5.5.40	A	5.5.41	A;C;D	5.5.42	8	5.5.43	A	5.5.44	A;C
5.5.45	B;C;D	5.5.46	50	5.5.47	C;D	5.5.48	A;B;D	5.5.49	Q-Q
5.5.50	B	5.5.51	A	5.5.52	B	5.5.53	B	5.5.54	A
5.5.55	D	5.5.56	C	5.6.1	C	5.7.1	B	5.7.2	B
5.7.3	A	5.7.4	C	5.7.5	4	5.7.6	C	5.7.7	A
5.7.8	A	5.7.9	D	5.7.10	A	5.7.11	B	5.7.12	C
5.8.1	B;C	5.8.2	B;D	5.9.1	N/A	5.9.2	N/A	5.9.3	3
5.9.4	C	5.9.5	A	5.9.6	C	5.9.7	C	5.9.8	C
5.9.9	C	5.9.10	4	5.9.11	698 : 698	5.9.12	6	5.9.13	A;B
5.10.1	A	5.10.2	A	5.10.3	A	5.10.4	C	5.10.5	B
5.10.6	A	5.10.7	A	5.11.1	C	5.12.1	C	5.12.2	A
5.12.3	C	5.13.1	N/A	5.14.1	B	5.14.2	C	5.14.3	A

5.14.4	0.00	5.14.5	D	5.15.1	N/A	5.15.2	N/A	5.15.3	N/A
5.15.4	N/A	5.15.5	N/A	5.15.6	N/A	5.15.7	D	5.15.8	N/A
5.15.9	B	5.15.10	C	5.15.11	D	5.15.12	C	5.15.13	N/A
5.15.14	A	5.15.15	D	5.15.16	B	5.15.17	D	5.15.18	A
5.15.19	D	5.15.20	D	5.15.21	4	5.15.22	C	5.15.23	1
5.15.24	C	5.15.25	A;B	5.15.26	2	5.15.27	B	5.15.28	1:1
5.15.29	C	5.15.30	C	5.15.31	B	5.16.1	N/A	5.16.2	N/A
5.16.3	D	5.16.4	C	5.16.5	C	5.16.6	C	5.16.7	B
5.16.8	C	5.16.9	C	5.16.10	A	5.16.11	A	5.16.12	D
5.16.13	D	5.16.14	D	5.16.15	C	5.17.1	A	5.17.2	B;C
5.18.1	N/A	5.18.2	N/A	5.18.3	N/A	5.18.4	N/A	5.18.5	N/A
5.18.6	N/A	5.18.7	D	5.18.8	N/A	5.18.9	N/A	5.18.10	A
5.18.11	C	5.18.12	N/A	5.18.13	C	5.18.14	N/A	5.18.15	N/A
5.18.16	N/A	5.18.17	C	5.18.18	D	5.18.19	D	5.18.20	C
5.18.21	B	5.18.22	C	5.18.23	A	5.18.24	X	5.18.25	A
5.18.26	C	5.18.27	A	5.18.28	C	5.18.29	C	5.18.30	B
5.18.31	D	5.18.32	B	5.18.33	C	5.18.34	D	5.18.35	2
5.18.36	A	5.18.37	2	5.18.38	2.6	5.18.39	7	5.18.40	D
5.18.41	5	5.18.42	A	5.18.43	819 : 820 ; 205 : 205	5.18.44	B	5.18.45	2
5.18.46	2	5.18.47	26:26	5.18.48	A;C;D	5.18.49	3:3	5.18.50	3
5.18.51	A;B	5.18.52	D	5.18.53	C	5.18.54	C	5.18.55	D
5.18.56	A	5.18.57	B	5.18.58	D	5.19.1	D	5.19.2	D
5.19.3	D	5.19.4	B	5.19.5	C	5.19.6	B	5.19.7	B
5.19.8	A	5.19.9	B	5.19.10	B	5.19.11	A	5.19.12	B
5.19.13	D	5.19.14	A	5.19.15	A	5.19.16	C	5.19.17	A
5.19.18	C	5.19.19	A	5.19.20	A	5.19.21	A;C;D	5.19.22	A
5.19.23	C	5.19.24	B;C	5.19.25	C	5.19.26	X	5.19.27	D
5.19.28	D	5.19.29	C	5.19.30	D	5.20.1	A		